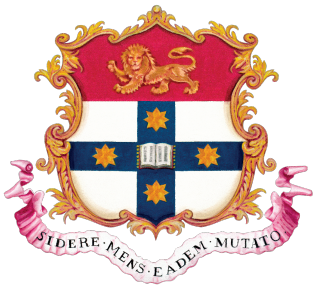
Graded Representation Theory of Equivariant Neural Networks

Mani Shayestehfar

A thesis submitted in partial fulfillment of
the requirements for the degree of
Master of Mathematical Sciences

Pure Mathematics
University of Sydney



November 2026

Statement of originality

This is to certify that the content of this thesis is my own work. This thesis has not been submitted for any other degree or purpose. I certify that the intellectual content of this thesis is the product of my own work, and that all assistance received in preparing this thesis and all sources have been acknowledged. The material of Chapters 2–7 also appear in the submitted preprint [Sha26a], and the architecture and experiment of Chapter 8 appear in the accepted paper [Sha26b].

Acknowledgements

This research was supervised by Dani Tubbenhauer, to whom I am sincerely grateful. Thank you for the countless discussions, for your patience, and for your unwavering guidance throughout. Your mentorship has shaped who I am as a mathematician over the past year, and I feel its growing influence every day.

I would like to thank Valeriano Aiello for his kind comments on [Sha26a], and I thank the anonymous reviewers of the NeurReps 2026 workshop for their constructive feedback on [Sha26b], as well as the reviewers of this thesis.

To my mother and father: thank you for bearing my absence with such grace, and for your constant love and support while I worked through this thesis. Everything I have achieved rests on the foundation you gave me. I am also deeply grateful to my close friends, whose conversations so often helped me see my research in a new light. In particular, I thank Daniel Bruwel, with whom I shared many inspiring discussions that shaped the work in Chapter 7.

Finally, to my dear partner, Eva: I could not have reached the finish line without you. You have stood beside me at every milestone, celebrating the good days and carrying me through the hard ones. I could not be prouder to share this life with you.

Use of AI

The initial core of this work was developed without AI assistance. Discussions with Claude Fable pointed towards sources such as [Jan97], where the connection between $L^2(\gamma_L)$ and $\mathrm{Sym}^k L$ was found. ChatGPT Sol assisted with polishing arguments and proofs, and with the literature review.

For the work of Chapter 8, generative AI tools also assisted with experiment design, code generation and debugging, and computational analysis, as declared in [Sha26b]. After using each tool, I reviewed and edited the content as needed, and I take full responsibility for the content of this thesis.

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Chapter 1

Introduction

A neural network computes a function by alternating two kinds of operation: *linear maps*, whose matrix entries are the trainable *weights*, and a fixed scalar nonlinearity $\sigma : \mathbb{R} \to \mathbb{R}$, the *activation function*, applied separately to each coordinate. Figure 1.1 shows a network with one hidden layer. Its input and output lie in real vector spaces V_{in} and V_{out}, and its hidden layer is $H = \mathbb{R}^{\mathscr{B}}$, whose coordinates (the *neurons*) are indexed by a finite set $\mathscr{B}$. The network computes

$$f(v) = R\,\sigma_{\mathscr{B}}(Av), \qquad \sigma_{\mathscr{B}}(z)_b = \sigma(z_b) \quad (b \in \mathscr{B}), \tag{1}$$

where the *lift* $A : V_{\text{in}} \to H$ and the *readout* $R : H \to V_{\text{out}}$ are linear, and a typical activation is $\mathrm{ReLU}(t) = \max(0, t)$. Deeper networks compose several such layers. Fixing the spaces and the activation fixes an *architecture*, which defines a family of functions as the weights vary.

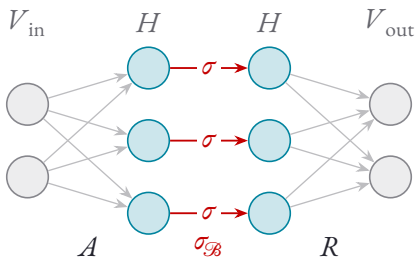


FIGURE 1.1. A neural network with one hidden layer, $f(v) = R\,\sigma_{\mathscr{B}}(Av)$.

When the data carry symmetries, one often expects the network to respect them: rotating, translating or relabelling an input should naturally have a corresponding effect on the output. Mathematically, the feature spaces are then representations of a group G, and the linear parts of the network are G-equivariant maps. This is precisely the viewpoint promoted by geometric deep learning [Bro+21].

Representation theory is very effective for understanding the linear maps. A finite-dimensional representation can be decomposed into irreducible pieces, and an equivariant linear map cannot mix two non-isomorphic irreducibles. In a basis adapted to the isotypic decomposition, every equivariant linear map is block diagonal.

The activation function changes the picture. It is nonlinear and is usually applied coordinatewise, so it can create interactions between irreducible pieces which are completely invisible to linear representation theory. The simplest example already occurs for the symmetric group S_3.

Let S_3 permute the coordinates of

$$\mathbb{R}^3 = L_0 \oplus L_1, \qquad L_0 = \mathrm{span}\{(1,1,1)\}, \qquad L_1 = \{x : x_1 + x_2 + x_3 = 0\}.$$

The first summand is the trivial representation and the second is the two-dimensional standard representation. There is no nonzero equivariant *linear* map from L_1 to L_0. Coordinatewise squaring, however, gives

$$p_0(x_1^2, x_2^2, x_3^2) = \frac{\|x\|^2}{3}(1,1,1), \qquad x \in L_1, \tag{2}$$

where p_0 is orthogonal projection onto L_0. In the notation of Figure 1.1, this is the network with input $V_{\text{in}} = L_1$, hidden layer $H = \mathbb{R}^3$ with its coordinate basis, lift the inclusion $L_1 \hookrightarrow \mathbb{R}^3$, activation $\sigma(t) = t^2$ and readout $R = p_0$. Thus a coordinatewise nonlinearity creates a quadratic interaction between two representation types which no linear equivariant map can connect.

This is the starting point of the thesis. We ask not merely whether a nonlinear equivariant interaction exists, but at which degrees it occurs and whether a given neural network layer actually produces it.

1.1. From existence to degree

Gibson, Tubbenhauer and Williamson (GTW) [GTW25] study the first, degree-free version of this question. For irreducible real representations L and K of a finite group, they show that a nonzero equivariant piecewise linear map $L \to K$ exists precisely when

$$\ker L \subseteq \ker K. \tag{3}$$

Here piecewise linear means continuous and affine on each of finitely many polyhedra covering L. This is already very different from the linear theory: nonlinear equivariant maps can connect many different irreducible representations.

Their "piecewise linear Schur's lemma" (Theorem 2.6.1), however, does not say how the interaction occurs. The interaction in (2), for example, is quadratic, whereas another interaction might first appear only in degree three or four. One would like to distinguish interactions by the degree in which they occur, and then to understand which of these degrees a network actually uses.

For polynomial maps there is an obvious notion of degree: the degree-k equivariant maps $L \to K$ correspond to

$$\mathrm{Hom}_G(\mathrm{Sym}^k L, K).$$

To analyse non-polynomial activations such as ReLU by degree, we extend this grading. We do so by placing the standard Gaussian measure on L. Just as monomials organise ordinary polynomials by degree, Hermite polynomials form an orthogonal basis for functions with respect to the Gaussian measure, and grouping them by total degree gives an orthogonal decomposition of the space of square-integrable functions, the *polynomial chaos* decomposition.

This grading is well known in the theory of Gaussian Hilbert spaces. Its usefulness here is that it gives a common language for polynomial and non-polynomial activations. A ReLU layer, for example, has infinitely many Hermite components, and we can ask exactly which representation-theoretic interactions occur in each one. In this language the GTW criterion becomes a statement about whether *some* degree is available, and the natural refinement asks *which* degrees are available and where the first of them lies.

1.2. What does one layer produce?

Knowing that a degree-k equivariant map $L \to K$ exists does not mean that a given neural network layer produces one. Let $L \subseteq V_{\text{in}}$ and $K \subseteq V_{\text{out}}$ be irreducible summands. Restricting the network (1) to inputs in L and projecting its output orthogonally onto K gives a *coordinatewise layer*

$$L \xrightarrow{A} H \xrightarrow{\sigma_{\mathcal{B}}} H \xrightarrow{R} K,$$

where, by abuse of notation, A now denotes the restriction of the lift to L, and R the readout followed by projection onto K. These are the blocks of the network with respect to irreducible decompositions of its input and output (Definition 2.6.2).

There are two separate reasons why a degree permitted by representation theory may fail to appear in such a layer: the linear maps A and R may not produce the relevant polynomial interaction, or the activation σ may have no component in that degree. Together with symmetry itself, this gives three separate questions:

(a) Does symmetry allow a degree-k equivariant map $L \to K$?

(b) Do the chosen linear maps A and R actually produce one?

(c) Does the activation σ retain degree k?

The central observation is that, in the Gaussian grading, these three tests can be separated explicitly. When G acts transitively on the hidden basis, the degree-k component is the chaos projection of a polynomial built from A and R, multiplied by a scalar depending only on σ and the common coordinate scale. The factorisation identifies which test prevents a permitted interaction from appearing. We give an example for S_4 in which symmetry permits a quadratic interaction but the coordinates never produce it, whatever the activation, and an example for C_8 in which the coordinates produce a cubic interaction but ReLU removes it, while a bias restores it. Distinguishing these three mechanisms is the main point of the thesis.

This separation also guides network design: it identifies which interactions a family of layers spans, which activations retain the required degrees, and which coordinate systems suffice. For the symmetric group acting on vertices, pairs and triples of a graph, the relevant polynomial spaces can be written down explicitly. In the final part of the thesis we use these ideas to build a graph network on unordered triples whose nonlinear computation is organised by irreducible source and target, and we test this organisation on a synthetic benchmark, addressing the question of GTW regarding the use case of interaction graphs in architecture design.

1.3. Relation to existing work and scope

Our starting point is GTW's Theorem 2.6.1, as recalled above. We refine this existence question by recording the polynomial degrees in which an interaction can occur, and then ask which of these interactions are produced by a specified coordinatewise layer and activation.

For compact groups, the structure theorem of Aiello [Aie26, Theorem 2.5 and Remark 2.6] places this finite group analysis in a broader context. Let $\mathscr{C}$ be a compact group with identity

component $\mathscr{C}^\circ$. Every equivariant piecewise linear map $F : L \to K$ splits as

$$F(v + w) = \psi(v) + Tw, \qquad v \in L^{\mathscr{C}^\circ}, \quad w \in (L^{\mathscr{C}^\circ})^\perp,$$

with $\psi : L^{\mathscr{C}^\circ} \to K^{\mathscr{C}^\circ}$ equivariant and piecewise linear and T equivariant and linear. Since $\mathscr{C}$ acts on the fixed subspaces $L^{\mathscr{C}^\circ}$ and $K^{\mathscr{C}^\circ}$ through a finite group, the nonlinear part reduces to a finite group action. Our degree criteria apply to ψ, and our layer criteria test its coordinatewise constructions. For connected groups, nonlinear dependence is confined to the trivial subrepresentations, which also explains why polynomial equivariants alone cannot detect piecewise linear existence for general compact groups.

At the level of layer construction, [Pac+24] classify the combinations of representations, coordinates and pointwise activations that give an equivariant activation map. We work with hidden permutation representations, where shared coordinatewise activations are equivariant, and analyse the interactions selected by the linear lift, activation and readout.

The analytic tools are classical. Gaussian chaos, Wick ordering and their interpretation through symmetric tensors are developed in [Jan97, Chapters 2–4], and Hermite expansions and the associated covariance identities also underlie the dual-activation framework of [DFS16]. We use these identities to separate the polynomial selected by an equivariant layer from the scalar coefficient supplied by its activation.

Our spanning construction connects with equivariant approximation theory: [Rav20] establishes universality using regular hidden actions, while [Yar22] obtains invariant and equivariant approximation through group averaging. Here regular representations provide an exact spanning construction in each polynomial degree, although matching finitely many Hermite components does not by itself control the approximation error.

For subset representations, the marked-orbit basis follows the orbit-sum principle used by [Pun+23] for equivariant graph polynomials, and the distinction between parameter sharing across dimensions and compatibility with embeddings already appears in [LD24]. Higher-order graph networks act on features indexed by tuples or subsets of vertices and use the full linear equivariant layer space on them [Mar+19; Mor+19]. Our graph architecture works with the same subset features, but organises the *nonlinear* computation by irreducible components, on top of a message-passing backbone [Xu+19].

1.4. Organisation

Chapters 2 to 4 establish the piecewise linear and Gaussian framework and the finite group degree criteria. Chapters 5 and 6 analyse coordinatewise layers, activation spectra and degreewise spanning constructions. Chapter 7 gives polynomial bases for subset features, and Chapter 8 applies the coordinatewise theory to graph triples. Chapter 9 contains conclusions and future directions. The experimental protocol and supplementary subset and projector calculations appear in Appendices A to C.

Code & media availability

The code used to generate the experiments in Chapter 8 is available in this GitHub repository. A summary video for [Sha26b] can be found here.

Chapter 2

Preliminaries and piecewise linear maps

This chapter fixes notation and states the standard results used later, with references for proofs [Ser77; FH91]. The final section reviews the piecewise linear representation theory of [GTW25].

2.1. Conventions

All representations are finite-dimensional and real unless otherwise stated, and are equipped with fixed invariant inner products. The acting group G is finite unless compact groups are explicitly allowed. For a representation L, we write $\ker L$ for the kernel of the action; it is a normal subgroup of G. We use the words *module* and *representation* interchangeably.

Every continuous representation L of a finite or compact group admits an invariant inner product, obtained by averaging an arbitrary inner product over the group (or integrating against Haar measure). With respect to it, G acts through the orthogonal group $O(L)$, and we call L an *orthogonal representation*. All adjoints are taken with respect to the fixed inner products; the adjoint of an equivariant linear map is again equivariant. We write $\mathrm{Hom}_G(L, K)$ for the equivariant linear maps $L \to K$.

A representation L is *faithful* if $\ker L = \{1\}$. If $N = \ker L$, then L is a faithful representation of $Q = G/N$. Conversely, representations of Q are identified, by *inflation* along $G \twoheadrightarrow Q$, with the G-representations on which N acts trivially, and $\mathrm{Hom}_Q = \mathrm{Hom}_G$ on these. For a subgroup $N \leq G$ we write K^N for the N-fixed subspace of K. If N is normal, then K^N is a subrepresentation.

2.2. Irreducible decompositions

The orthogonal complement of a subrepresentation of an orthogonal representation is again a subrepresentation, so every representation is an orthogonal direct sum of irreducibles (Maschke's theorem). We use Schur's lemma in the following real form [Ser77, Sections 2.2 and 13.2].

Proposition 2.2.1 (Schur's lemma). *Let L and K be irreducible real representations. If $L \not\cong K$, then $\mathrm{Hom}_G(L, K) = 0$. Otherwise, $\mathrm{End}_G(L)$ is a real division algebra, isomorphic to $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

L is *absolutely irreducible* if $\mathrm{End}_G(L) \cong \mathbb{R}$. Every irreducible representation of a symmetric group is absolutely irreducible, whereas the two-dimensional rotation representations of cyclic groups (Example 2.3.2) have $\mathrm{End}_G \cong \mathbb{C}$.

For an irreducible K, the *K-isotypic component* of L is the sum of all subrepresentations isomorphic to K. The isotypic decomposition of L is canonical, orthogonal for every invariant inner product, and preserved by equivariant linear maps; a decomposition into irreducibles is unique only up to

the choice of summands within each isotypic component. We write $[L : K]$ for the multiplicity of K in L, and call L *multiplicity-free* if every multiplicity is at most one.

Equivariant linear maps are detected by characters. For the group action $\rho_L : G \to \mathrm{GL}(L)$, the character of a real representation is real-valued and satisfies $\chi_L(g^{-1}) = \chi_L(g)$. For finite G,

$$\dim \mathrm{Hom}_G(L, K) = \frac{1}{|G|} \sum_{g \in G} \chi_L(g)\chi_K(g), \tag{4}$$

and when K is irreducible this equals $[L : K] \dim_{\mathbb{R}} \mathrm{End}_G(K)$. Since both sides of (4) are unchanged by complexification, $\mathrm{Hom}_G(L, K) \neq 0$ if and only if $L_{\mathbb{C}} = L \otimes_{\mathbb{R}} \mathbb{C}$ and $K_{\mathbb{C}}$ share a complex irreducible constituent.

2.3. Permutation representations

For a finite G-set X, the *permutation representation* $\mathbb{R}[X]$ has orthonormal basis $\{e_x : x \in X\}$ and action $g e_x = e_{gx}$. More generally, we say that H is a permutation representation *with basis* $\mathcal{B}$ if $\mathcal{B}$ is an orthonormal basis of H permuted by G, and we identify $H = \mathbb{R}^{\mathcal{B}}$, writing $z = \sum_b z_b b$, so that $(gz)_b = z_{g^{-1}b}$. The character of $\mathbb{R}[X]$ counts fixed points, $\chi_{\mathbb{R}[X]}(g) = |X^g|$, and $\dim \mathbb{R}[X]^G$ is the number of G-orbits on X.

The following elementary observation is the reason coordinatewise activations are compatible with symmetry.

Lemma 2.3.1. *If H is a permutation representation with basis $\mathcal{B}$ and $f : \mathbb{R} \to \mathbb{R}$ is any function, then $f_{\mathcal{B}}(z) = \sum_b f(z_b) b$ is equivariant.*

Proof. $(f_{\mathcal{B}}(gz))_b = f(z_{g^{-1}b}) = (f_{\mathcal{B}}(z))_{g^{-1}b} = (g f_{\mathcal{B}}(z))_b$. □

The permutation-basis hypothesis guarantees equivariance for every scalar function; see [Pac+24; GTW25] for the general question of when coordinatewise application is equivariant. A transitive G-set with point stabiliser $H \leq G$ gives $\mathbb{R}[G/H] = \mathrm{Ind}_H^G \mathbf{1}$, and for $H = \{1\}$ we obtain the *regular representation* $\mathbb{R}[G]$. Frobenius reciprocity takes the form $\mathrm{Hom}_G(\mathrm{Ind}_H^G \mathbf{1}, K) \cong K^H$, the isomorphism sending T to $T(e_H)$, where e_H is the basis vector of the trivial coset [Ser77].

Example 2.3.2 (Cyclic groups). Let $C_n = \langle a \rangle$. Its complex irreducible representations are the characters $W_t : a \mapsto \zeta^t$, $\zeta = e^{2\pi i/n}$, $t \in \mathbb{Z}/n\mathbb{Z}$. Its real irreducible representations are the trivial representation $\mathbf{1} = W_0$; for even n, the sign character $\varepsilon = W_{n/2}$; and, for each pair $\pm t$ with $2t \not\equiv 0 \pmod n$, the rotation representation $\rho_t \cong \rho_{-t}$ on $\mathbb{R}^2$ in which a acts by rotation through $2\pi t/n$. Here $(\rho_t)_{\mathbb{C}} \cong W_t \oplus W_{-t}$, and $\ker \rho_t$ is the subgroup of order $\gcd(t, n)$. The regular representation is multiplicity-free:

$$\mathbb{R}[C_n] \cong \mathbf{1} \oplus \varepsilon \oplus \bigoplus_{0 < t < n/2} \rho_t,$$

with ε present only for even n.

Example 2.3.3 (Symmetric groups). The irreducible real representations of S_n are the Specht modules S^λ, $\lambda \vdash n$, all absolutely irreducible [Jam78; Sag01]. The permutation representation $\mathbb{R}^n$

is the sum of the trivial representation $S^{(n)}$, spanned by $(1,\dots,1)$, and the *standard representation* $S^{(n-1,1)} = \{x : \sum_i x_i = 0\}$. For S_3, with conjugacy classes of sizes $1, 3, 2$, the character table of the trivial, standard, and sign representations is

	e	(12)	(123)
$\mathbf{1} = S^{(3)}$	1	1	1
$\mathrm{std} = S^{(2,1)}$	2	0	-1
$\varepsilon = S^{(1,1,1)}$	1	-1	1

2.4. Polynomial maps and symmetric powers

Polynomial degree is the basic grading used throughout the thesis. We begin by recalling the standard identification between homogeneous polynomial maps and maps out of symmetric powers.

Definition 2.4.1. For a real representation L and $k \geq 1$, define the *kth symmetric power*

$$\mathrm{Sym}^k L = L^{\otimes k} \Big/ \operatorname{span}\{t - \pi t : t \in L^{\otimes k},\ \pi \in S_k\}, \qquad \mathrm{Sym}^0 L = \mathbb{R},$$

where S_k acts by permuting tensor factors. The diagonal action of G on $L^{\otimes k}$ commutes with this action, and therefore descends to $\mathrm{Sym}^k L$.

For $x \in L$, we write x^k for the class of $x^{\otimes k}$ in $\mathrm{Sym}^k L$. Elements of this form are called *pure powers*, which span $\mathrm{Sym}^k L$, and the induced inner product is characterised by

$$\left\langle x^k, y^k \right\rangle = \left\langle x, y \right\rangle^k.$$

To avoid confusion, coordinatewise powers in a based permutation representation will instead be denoted by $x^{\odot k}$.

Definition 2.4.2. For real representations L and K, let $\mathrm{Poly}_k(L,K)$ denote the *space of homogeneous polynomial maps* $P : L \to K$ of degree k. The G-action is $(g \cdot P)(x) = gP(g^{-1}x)$, so that

$$\mathrm{Poly}_k(L,K)^G = \{P \in \mathrm{Poly}_k(L,K) : P(gx) = gP(x) \text{ for all } g \in G,\ x \in L\}.$$

Similarly, G acts on $\mathrm{Hom}(\mathrm{Sym}^k L, K)$ by $(g \cdot T)(y) = gT(g^{-1}y)$.

Proposition 2.4.3. *For every $k \geq 0$, the map $T \mapsto \big(x \mapsto T(x^k)\big)$ is an isomorphism*

$$\mathrm{Hom}_G(\mathrm{Sym}^k L, K) \xrightarrow{\cong} \mathrm{Poly}_k(L,K)^G.$$

Proof. Polarisation identifies a homogeneous polynomial map P of degree k with a symmetric k-linear map $\widetilde{P}$ satisfying $P(x) = \widetilde{P}(x,\dots,x)$, and the universal property of $\mathrm{Sym}^k L$ converts $\widetilde{P}$ into a unique linear map $T_P : \mathrm{Sym}^k L \to K$ with $T_P(x^k) = P(x)$. These constructions are mutually inverse and natural in L and K, so P is equivariant if and only if T_P is. □

We use this identification throughout to regard homogeneous degree-k equivariant polynomial maps as elements of $\mathrm{Hom}_G(\mathrm{Sym}^k L, K)$, so that their existence becomes a character computation via (4). For this, note that if $\rho_L(g)$ has complex eigenvalues $\lambda_1,\dots,\lambda_m$, then $\chi_{\mathrm{Sym}^k L}(g) = \sum_{|\alpha|=k} \lambda^\alpha$

over multi-indices $\alpha \in \mathbb{N}_0^m$ with $|\alpha| = k$; for instance,

$$\chi_{\mathrm{Sym}^2 L}(g) = \tfrac{1}{2}\left(\chi_L(g)^2 + \chi_L(g^2)\right). \tag{5}$$

Example 2.4.4. Let L_1 be the standard representation of S_3. By (5), $\chi_{\mathrm{Sym}^2 L_1}$ takes the values $3, 1, 0$ on e, (12), (123), so $\mathrm{Sym}^2 L_1 \cong \mathbf{1} \oplus L_1$. Thus, up to scaling, there is exactly one invariant quadratic form, $\|x\|^2$, and exactly one equivariant quadratic map $L_1 \to L_1$, namely $x \mapsto p_{L_1}(x^{\odot 2})$, where $x^{\odot 2} = (x_1^2, x_2^2, x_3^2)$ and p_{L_1} is orthogonal projection; it is nonzero, since it sends $(1, -1, 0)$ to $(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3})$.

2.5. Piecewise linear maps

Our nonlinear layers are piecewise linear, while the degree decomposition below lives in a Gaussian L^2 space. We first fix the class of maps to which both viewpoints apply.

Definition 2.5.1. A map $F : L \to K$ is *piecewise linear* if it is continuous and there is a finite covering of L by polyhedra on each of which F is affine.

We write $\mathrm{Hom}_G^{\mathrm{PL}}(L, K)$ for the equivariant ones. We follow the affine-on-pieces convention of [GTW25], which includes constant maps.

Sums and compositions of piecewise linear maps are piecewise linear (intersect the pieces), and so is a coordinatewise map $\sigma_{\mathcal{B}}$ when $\sigma : \mathbb{R} \to \mathbb{R}$ is continuous and piecewise linear with finitely many knots. Consequently every layer $x \mapsto R\sigma_{\mathcal{B}}(Ax)$ considered in this thesis, with A, R linear and σ such as ReLU, is an equivariant piecewise linear map, and so is any finite composition of such layers.

2.6. Piecewise linear representation theory

The example from (2) shows that replacing squaring by the absolute value gives a nonzero equivariant piecewise linear map between non-isomorphic irreducibles. Hence Schur's lemma maximally fails in this setting. Its replacement is the following.

Theorem 2.6.1 (Piecewise linear Schur's lemma [GTW25, Theorem 2H.3]). *Let G be finite and let L, K be irreducible real representations. Then*

$$\mathrm{Hom}_G^{\mathrm{PL}}(L, K) \neq 0 \quad \Longleftrightarrow \quad \ker L \subseteq \ker K.$$

The forward implication holds because an equivariant map takes values in the subrepresentation $K^{\ker L}$, and since $K^{\ker L}$ is a nonzero subrepresentation of the irreducible K, it equals K; the converse is proved in [GTW25] by averaging a piecewise linear bump function over a free orbit. Theorem 4.2.3 proves the corresponding polynomial existence criterion, with a degree bound.

2.6.1. Interaction graphs

Let $V = \bigoplus_{i=1}^r L_i$ be a fixed orthogonal decomposition into irreducibles, with inclusions ι_i and orthogonal projections p_i. The *blocks* of a map $F : V \to V$ are $F_i^j = p_j \circ F \circ \iota_i : L_i \to L_j$. For

nonlinear F the blocks do not determine F, but they record whether a signal in L_i produces a response in L_j.

Definition 2.6.2. The *interaction graph* Γ_F of an equivariant map $F : V \to V$ has vertices $L_1, \dots, L_r$ and an edge $L_i \to L_j$ whenever $F_i^j \not\equiv 0$. The *kernel-inclusion graph* $\Gamma_{\ker}$ has an edge $L_i \to L_j$ whenever $\ker L_i \subseteq \ker L_j$.

Each block of an equivariant piecewise linear map is again equivariant and piecewise linear, so Theorem 2.6.1 gives $\Gamma_F \subseteq \Gamma_{\ker}$ for every such F. If the vertices are instead the isotypic components, the resulting graph does not depend on any choice. Since all irreducibles in one isotypic component have the same kernel, the results below hold for this variant too.

2.6.2. Global realisability

The inclusion $\Gamma_F \subseteq \Gamma_{\ker}$ constrains each pair of summands. Every graph satisfying these pairwise constraints is realised by some equivariant piecewise linear map.

Theorem 2.6.3. *Let G be finite. A directed graph Γ on $L_1, \dots, L_r$ is the interaction graph of some equivariant piecewise linear $F : V \to V$ if and only if $\Gamma \subseteq \Gamma_{\ker}$. Moreover F can be chosen with $F(0) = 0$.*

Proof. "$\Rightarrow$" was noted above. For "$\Leftarrow$" we first show that for each edge $L_i \to L_j$ of $\Gamma_{\ker}$ there is a nonzero equivariant piecewise linear $\phi_i^j : L_i \to L_j$ with $\phi_i^j(0) = 0$. If L_j is nontrivial, take any nonzero map from Theorem 2.6.1; then $\phi_i^j(0) \in L_j^G = 0$. If $L_j \cong \mathbb{R}$ is trivial, take $\phi_i^j(x) = \max_{g \in G} \left|\ell(gx)\right|$ for some nonzero $\ell \in L_i^*$, which is invariant, piecewise linear and vanishes at 0.

Now put $\phi_i^j = 0$ for non-edges of Γ and $F = \sum_{i,j} \iota_j \circ \phi_i^j \circ p_i$. Then F is equivariant and piecewise linear with $F(0) = 0$, and for $x \in L_a$ we have $p_b F \iota_a(x) = \sum_i \phi_i^b(\delta_{ia} x) = \phi_a^b(x)$, because $\phi_i^b(0) = 0$ for $i \neq a$. Hence $F_a^b = \phi_a^b$, and $\Gamma_F = \Gamma$. □

Thus the kernel criterion also determines which interaction graphs can be realised by unrestricted piecewise linear maps. The questions of this thesis arise once one asks *how* an interaction is created, and whether a layer of a fixed architecture, such as a coordinatewise ReLU layer, creates it.

Chapter 3

Gaussian spaces and polynomial chaos

The activation functions used in practice, such as ReLU, are not polynomial, and a piecewise linear map has no degree in the usual sense. To speak about the degree in which a nonlinear equivariant map creates an interaction, we need a grading on a space of functions large enough to contain piecewise linear maps. This chapter introduces such a grading, the *polynomial chaos* (or Wiener chaos) decomposition of the Gaussian L^2 space, and proves the properties we need. The standard reference is [Jan97, Chapters 2–4].

3.1. Why a Gaussian measure

A grading of a function space by degree requires an inner product, so that "the degree-k part" of a function can be defined by orthogonal projection. Three requirements make the Gaussian measure the natural reference measure.

(a) Invariance. The grading should respect every symmetry we might impose. If the inner product is invariant under the full orthogonal group $O(L)$, then so is the grading, and hence it is automatically compatible with every orthogonal representation of every group. The standard Gaussian measure is $O(L)$-invariant.

(b) Polynomials. The graded pieces should be built from polynomials, so that the grading extends ordinary polynomial degree. This requires all moments to be finite and polynomials to be dense. Both hold for the Gaussian measure.

(c) Independence. In an orthonormal basis the coordinates of a standard Gaussian vector are independent, so multivariable questions reduce to one-variable ones. The Herschel–Maxwell theorem states that, in dimension at least two, a rotation-invariant probability measure with independent coordinates is a centred Gaussian [Bry95].

The Gaussian measure is used purely as a reference measure defining the grading. It is not an assumption about the distribution of data or of trained features, and all statements of the form "$F = 0$" that we derive from it are statements about continuous maps, which hold pointwise (see Proposition 3.2.3 and Remark 3.2.4).

3.2. Gaussian spaces

Throughout, L and K are orthogonal representations and $m = \dim L$.

Definition 3.2.1. The *standard Gaussian measure* γ_L on L is the probability measure

$$d\gamma_L(x) = (2\pi)^{-m/2} e^{-\|x\|^2/2}\, dx,$$

where dx is the Lebesgue measure in any orthonormal basis of L. We write $X \sim \gamma_L$ for a random vector with this distribution.

Since γ_L does not depend on the chosen basis, it is invariant under $O(L)$, and hence under G. It has moments of every order, and for every $a \in L$ the linear form $\langle X, a\rangle$ is centred normal with variance $\|a\|^2$.

Definition 3.2.2. The *Gaussian L^2 space* $L^2(\gamma_L; K)$ is the Hilbert space of maps $F : L \to K$ with $\mathbb{E}\,\|F(X)\|^2 < \infty$, modulo γ_L-null sets, with inner product

$$\langle F, F'\rangle = \mathbb{E}\,\langle F(X), F'(X)\rangle_K .$$

The group G acts on it by $(g\cdot F)(x) = gF(g^{-1}x)$. We write $L^2(\gamma_L) = L^2(\gamma_L; \mathbb{R})$, so that $L^2(\gamma_L; K) = L^2(\gamma_L) \otimes K$.

The action of G on $L^2(\gamma_L; K)$ is unitary, because γ_L is G-invariant and G acts orthogonally on K. Its fixed subspace $L^2(\gamma_L; K)^G$ consists of the (classes of) equivariant square-integrable maps. The Hermite expansion of Sections 3.3 and 3.4 gives an orthogonal grading of this space.

All piecewise linear maps, and hence all layers considered in this thesis, lie in this space.

Proposition 3.2.3. *Every piecewise linear map $F : L \to K$ lies in $L^2(\gamma_L; K)$.*

Proof. Since F is affine on finitely many polyhedral pieces, there exist constants $C, D \geq 0$ such that $\|F(x)\| \leq C\,\|x\| + D$ for all $x \in L$. Hence $\|F(x)\|^2 \leq 2C^2\,\|x\|^2 + 2D^2$. If $X \sim \gamma_L$, then $\mathbb{E}\,\|X\|^2 < \infty$, so

$$\int_L \|F(x)\|^2\, d\gamma_L(x) \leq 2C^2\mathbb{E}\,\|X\|^2 + 2D^2 < \infty.$$

Thus $F \in L^2(\gamma_L; K)$. □

Remark 3.2.4. Two continuous maps agreeing γ_L-almost everywhere agree everywhere, because γ_L has full support and the set where they differ is open. Consequently, the chaos components determine a continuous map uniquely.

3.3. Hermite polynomials

We use the probabilists' Hermite polynomials He_k, defined by the generating function $\exp(tx - t^2/2) = \sum_{k\geq 0} \mathrm{He}_k(x)t^k/k!$ and normalised so that $\mathbb{E}[\mathrm{He}_j(Z)\mathrm{He}_k(Z)] = k!\,\delta_{jk}$ for $Z \sim N(0,1)$.

We write $\varphi(t) = (2\pi)^{-1/2}e^{-t^2/2}$ for the standard normal density and $Z \sim N(0,1)$. The first Hermite polynomials are $\mathrm{He}_0 = 1$, $\mathrm{He}_1 = x$, $\mathrm{He}_2 = x^2 - 1$, $\mathrm{He}_3 = x^3 - 3x$ and $\mathrm{He}_4 = x^4 - 6x^2 + 3$. The following properties all follow from the generating function [Jan97, Section 3].

Lemma 3.3.1. *For all $k \geq 0$:*

(a) $\mathrm{He}_k(x)\varphi(x) = (-1)^k \varphi^{(k)}(x)$, *and* He_k *is monic of degree* k *with the parity of* k.

(b) $\mathrm{He}_{k+1}(x) = x\mathrm{He}_k(x) - k\mathrm{He}_{k-1}(x)$ *and* $\mathrm{He}_k' = k\mathrm{He}_{k-1}$, *with* $\mathrm{He}_{-1} := 0$.

(c) $(\mathrm{He}_k\varphi)' = -\mathrm{He}_{k+1}\varphi$.

(d) $\mathrm{He}_{2m}(0) = (-1)^m \frac{(2m)!}{2^m m!}$ *and* $\mathrm{He}_{2m+1}(0) = 0$.

(e) $\mathbb{E}[\mathrm{He}_j(Z)\mathrm{He}_k(Z)] = k!\,\delta_{jk}$.

Proposition 3.3.2 ([Jan97, Theorem 2.6]). *The polynomials* $\mathrm{He}_k/\sqrt{k!}$ *form an orthonormal basis of* $L^2(\gamma_{\mathbb{R}})$. *Hence every* $f \in L^2(\gamma_{\mathbb{R}})$ *has an* L^2*-convergent expansion* $f = \sum_k a_k \mathrm{He}_k$ *with* $a_k = \frac{1}{k!}\mathbb{E}[f(Z)\mathrm{He}_k(Z)]$.

3.4. Polynomial chaos

In orthonormal coordinates $x_1, \dots, x_m$ on L, the products $\prod_i \mathrm{He}_{k_i}(x_i)$ with $\sum_i k_i = k$ span the kth *polynomial chaos* $\mathscr{H}_k(L)$, and

$$L^2(\gamma_L) = \widehat{\bigoplus}_{k\geq 0} \mathscr{H}_k(L), \tag{6}$$

where $\widehat{\oplus}$ is the Hilbert direct sum, consisting of sequences of vectors whose squared norms are summable. Thus the chaos index is a polynomial degree after orthogonal removal of lower-degree terms. For example, $\mathrm{He}_2(x) = x^2 - 1$ lies entirely in the second chaos. We write Π_k for the orthogonal projection onto $\mathscr{H}_k(L)$ and use the same symbol for the induced projection on K-valued maps:

$$\Pi_k : L^2(\gamma_L; K) \longrightarrow \mathscr{H}_k(L) \otimes K.$$

The chaos $\mathscr{H}_k(L)$ is defined using a basis, but it does not depend on the choice of orthonormal basis. This is what makes it compatible with symmetry.

Lemma 3.4.1. *Let* $\mathscr{P}_{\leq k}(L)$ *denote the polynomial functions on* L *of degree at most* k. *Set* $\mathscr{P}_{\leq -1}(L) = \{0\}$ *and take orthogonal complements in* $L^2(\gamma_L)$. *Then*

$$\mathscr{H}_k(L) = \mathscr{P}_{\leq k}(L) \cap \mathscr{P}_{\leq k-1}(L)^{\perp} \cong \mathscr{P}_{\leq k}(L)/\mathscr{P}_{\leq k-1}(L).$$

In particular, Π_k *commutes with* $F \mapsto F \circ u^{-1}$ *for every* $u \in O(L)$.

Proof. Since each He_j is monic of degree j, the products $\prod_i \mathrm{He}_{k_i}(x_i)$ with $\sum_i k_i \leq k$ span $\mathscr{P}_{\leq k}(L)$, so $\mathscr{P}_{\leq k}(L) = \bigoplus_{j\leq k} \mathscr{H}_j(L)$ orthogonally. The spaces $\mathscr{P}_{\leq j}(L)$ are coordinate-free and preserved by the unitary operators $F \mapsto F \circ u^{-1}$, hence so is the orthogonal complement of one in the next. □

For K-valued maps, Π_k acts on the scalar factor of $L^2(\gamma_L) \otimes K$, so it also commutes with linear maps applied to the target.

Lemma 3.4.2. *Π_k restricts to an $O(L)$-equivariant isomorphism from the homogeneous polynomials of degree k onto $\mathscr{H}_k(L)$, and $P - \Pi_k P$ has degree at most $k-1$ for such P.*

Proof. On $\mathscr{P}_{\le k} = \mathscr{P}_{\le k-1} \oplus \{\text{homogeneous of degree } k\}$, the projection Π_k has kernel $\mathscr{P}_{\le k-1}$ by Lemma 3.4.1; dimensions then match. □

Composing with the identification of homogeneous polynomials with $\mathrm{Sym}^k(L^*)$, and $L^* \cong L$, gives the $O(L)$-equivariant identification $\mathscr{H}_k(L) \cong \mathrm{Sym}^k(L^*) \cong \mathrm{Sym}^k L$ [Jan97]. Thus $\mathscr{H}_k(L)$ consists of polynomials of degree at most k orthogonal to all lower-degree polynomials, rather than of homogeneous polynomials.

Lemma 3.4.3. *If $\ell(x) = \langle x, a\rangle$ with $\|a\| = s > 0$, then $\Pi_k(\ell^k) = s^k \mathrm{He}_k(\ell/s)$.*

Proof. In an orthonormal basis with $e_1 = a/s$, the function $s^k \mathrm{He}_k(x_1)$ lies in $\mathscr{H}_k(L)$ and differs from $\ell^k = s^k x_1^k$ by a polynomial of degree less than k. □

Inner products of projected powers are computed by Mehler's formula.

Lemma 3.4.4 (Hermite correlation identity, from Mehler's formula [Jan97, Theorem 3.9]). *Let (X, Y) be jointly standard normal with correlation ρ. Then*

$$\mathbb{E}[\mathrm{He}_a(X)\mathrm{He}_b(Y)] = a!\,\rho^a\,\delta_{ab}.$$

Proof. This is a standard identity. See [Jan97, Theorem 3.9]. □

Together with Lemma 3.4.3, Mehler's formula computes inner products of projected powers of linear forms: for nonzero $a, b \in L$ and $\ell_a = \langle\cdot, a\rangle$, $\ell_b = \langle\cdot, b\rangle$, the variables $\ell_a(X)/\|a\|$ and $\ell_b(X)/\|b\|$ are jointly standard normal with correlation $\langle a, b\rangle / (\|a\|\,\|b\|)$, so

$$\left\langle \Pi_k(\ell_a^k), \Pi_k(\ell_b^k) \right\rangle_{L^2(\gamma_L)} = k!\,\langle a, b\rangle^k . \tag{7}$$

Since $\left\langle a^k, b^k \right\rangle_{\mathrm{Sym}^k L} = \langle a, b\rangle^k$, the map $a^k \mapsto \Pi_k(\ell_a^k)$ is $\sqrt{k!}$ times an isometry $\mathrm{Sym}^k L \to \mathscr{H}_k(L)$. This is the concrete form of the identification above, and (7) underlies all Gram formulae in Chapter 5 and Section 6.4.

Chapter 4

Degree-resolved equivariant maps

We first identify the equivariant part of each polynomial chaos. We then relate occurrence at some degree to the finite group existence criterion.

4.1. The equivariant decomposition

The Gaussian grading is useful here only if it respects equivariance. The next theorem says that it does so, degree by degree.

Theorem 4.1.1. *Let G be a finite or compact group, and let L, K be finite-dimensional continuous orthogonal real G-representations. Then*

$$L^2(\gamma_L; K)^G = \widehat{\bigoplus}_{k\geq 0} (\mathscr{H}_k(L) \otimes K)^G,$$

with canonical linear isomorphisms

$$(\mathscr{H}_k(L) \otimes K)^G \cong \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$$

for every $k \geq 0$.

Proof. Tensoring (6) with K gives

$$L^2(\gamma_L; K) = \widehat{\bigoplus}_{k\geq 0} (\mathscr{H}_k(L) \otimes K).$$

Since each chaos projection is G-equivariant, taking fixed points yields the first decomposition. The second follows from Proposition 2.4.3 and the equivariant identification $\mathrm{Poly}_k(L, K) \cong \mathscr{H}_k(L) \otimes K$ induced by Π_k. Explicitly, $T \in \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$ corresponds to $\Pi_k P_T$, where $P_T(x) = T(x^k)$. By (7) and the induced inner product on $\mathrm{Sym}^k L$,

$$\|\Pi_k P_T\|_{L^2}^2 = k!\, \|T\|_{\mathrm{HS}}^2,$$

where $\|\cdot\|_{\mathrm{HS}}$ denotes the Hilbert–Schmidt norm. Thus the identification is isometric when the degree-k Hom space is equipped with the inner product scaled by $k!$. □

Corollary 4.1.2. *Every equivariant piecewise linear map $F : L \to K$ admits a unique orthogonal decomposition*

$$F = \sum_{k\geq 0} \Pi_k F, \qquad \Pi_k F \in (\mathscr{H}_k(L) \otimes K)^G,$$

with convergence in $L^2(\gamma_L; K)$. Equivalently, each component is uniquely represented by an element $T_k \in \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$. The Gaussian-chaos transform, $F \mapsto (\Pi_k F)_{k\geq 0}$, is injective on continuous square-integrable equivariant maps.

Proof. Square-integrability follows from Proposition 3.2.3, and the decomposition and the representatives T_k follow from Theorem 4.1.1. Orthogonality gives uniqueness in L^2, and full support of the Gaussian measure gives pointwise uniqueness for continuous maps. □

Corollary 4.1.2 establishes injectivity but does not characterise the image of the transform. In particular, not every square-summable sequence of equivariant polynomial components arises from a piecewise linear map (for instance $x \mapsto \|x\|^2 - m$, which lies in a single chaos, is not piecewise linear). The finite group existence questions considered here do not require such a characterisation. The distinction becomes essential for compact connected groups, as discussed in Chapter 9.

4.2. Degree sets and detection

We now forget multiplicities and record only the degrees in which the target can occur. This is the information needed for the existence and parity questions below.

Definition 4.2.1. For representations L, K define the *degree set*

$$D(L,K) := \{k \geq 0 : \mathrm{Hom}_G(\mathrm{Sym}^k L, K) \neq 0\},$$

and $d(L,K) := \min D(L,K) \in \mathbb{N}_0 \cup \{\infty\}$. We use $\min \emptyset = \infty$.

If $K^G \neq 0$, then degree zero records constant equivariant maps. When the question is transfer of input information, the first positive degree may therefore be the more relevant quantity.

Proposition 4.2.2. *Let $L \neq 0$, and let G be finite or compact. If $k \in D(L,K)$, then $k + 2m \in D(L,K)$ for every $m \in \mathbb{N}_0$. Consequently, for each $\varepsilon \in \{0,1\}$ define*

$$d_\varepsilon(L,K) := \min\{k \in D(L,K) : k \equiv \varepsilon \pmod 2\}.$$

Then

$$D(L,K) = (d_0(L,K) + 2\mathbb{N}_0) \cup (d_1(L,K) + 2\mathbb{N}_0)\,,$$

where a term is omitted if $d_\varepsilon(L,K)$ is infinite.

Proof. Since G is finite or compact, L admits a G-invariant inner product, so $q(x) = \|x\|^2$ is a nonzero invariant quadratic polynomial. Hence multiplication by q sends

$$\mathrm{Poly}_k(L,K)^G \longrightarrow \mathrm{Poly}_{k+2}(L,K)^G.$$

This map is injective because the polynomial ring $\mathbb{R}[L]$ is an integral domain and $q \neq 0$. Hence occurrence exists in increments of two, and each parity is empty or consists of all degrees of that parity from its first occurrence onwards. □

Thus, once the first even and odd occurrences are known, the entire degree set is known. The remaining task is to determine those two thresholds.

Theorem 4.2.3 (Finite group degree detection). *Let $L \neq 0$ and K be (not necessarily irreducible) real G-representations, and put $N = \ker L$ and $Q = G/N$. Then*

$$D(L,K) \neq \emptyset \quad \Longleftrightarrow \quad K^N \neq 0.$$

In this case $d(L,K) \leq |Q| - 1$. If K is irreducible, the condition is equivalent to $\ker L \subseteq \ker K$.

Proof. Suppose $k \in D(L,K)$ and choose $0 \neq T \in \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$. Since $N = \ker L$ acts trivially on $\mathrm{Sym}^k L$, we have $nT(y) = T(ny) = T(y)$ for all $n \in N$ and $y \in \mathrm{Sym}^k L$. Thus $\mathrm{im}\, T \subseteq K^N$, and since $T \neq 0$ we obtain $K^N \neq 0$. If K is irreducible, then K^N is G-stable because $N \lhd G$. Hence $K^N \neq 0$ implies $K^N = K$. Equivalently, $\ker L \subseteq \ker K$.

For the converse, assume $K^N \neq 0$. Both L and K^N descend to $Q = G/N$, and L is faithful as a Q-representation. We argue in three steps.

A free orbit exists. For $1 \neq q \in Q$ the fixed space L^q is a proper subspace, since L is faithful. The finite union of these proper subspaces cannot cover L, so some $v \in L$ has trivial stabiliser, and the orbit Qv consists of $|Q|$ distinct points.

The orbit can be separated by a linear form. A functional $\ell \in L^*$ fails to separate two distinct orbit points qv and $q'v$ precisely when $\ell(qv - q'v) = 0$. Since $qv - q'v \neq 0$, the set of such functionals is a hyperplane of L^*. There are only finitely many pairs of orbit points, so a finite union of these hyperplanes cannot cover L^*. Hence some $\ell \in L^*$ separates all points of Qv.

Interpolate. With such an ℓ, the Lagrange polynomial

$$p(x) = \prod_{1 \neq q \in Q} \frac{\ell(x) - \ell(qv)}{\ell(v) - \ell(qv)}$$

has degree at most $|Q| - 1$, with $p(v) = 1$ and $p(qv) = 0$. Fix $0 \neq w \in K^N$ and average:

$$F(x) = \sum_{q \in Q} p(q^{-1}x)\, qw.$$

Reindexing $q = aq'$ shows $F(ax) = aF(x)$, so F is equivariant, and $F(v) = w \neq 0$, so $F \neq 0$. The action is linear, so each homogeneous component F_j of F is separately equivariant, and some $F_j \neq 0$ with $j \leq |Q| - 1$. Thus $\mathrm{Poly}_j(L, K^N)^Q \neq 0$, and Proposition 2.4.3 gives $\mathrm{Hom}_Q(\mathrm{Sym}^j L, K^N) \neq 0$. Inflating along $G \twoheadrightarrow Q$ and composing with the inclusion $K^N \hookrightarrow K$ gives $0 \neq \mathrm{Hom}_G(\mathrm{Sym}^j L, K^N) \subseteq \mathrm{Hom}_G(\mathrm{Sym}^j L, K)$, so $j \in D(L,K)$. □

The key point here is that every irreducible representation of a finite group occurs somewhere in the symmetric algebra of a faithful representation. This finite-orbit interpolation argument does not extend to general compact connected groups.

Remark 4.2.4 (Compact connected groups). Theorems 2.6.1 and 4.2.3 show that, for finite groups and irreducible representations, piecewise linear and polynomial existence coincide. This fails for compact connected groups. Let $G = SO(2)$ act on $L = \mathbb{C}$ by $e^{i\theta} \cdot z = e^{i\theta} z$ and on $K = \mathbb{C}$ by $e^{i\theta} \cdot w = e^{2i\theta} w$. Then $\ker L = \{1\} \subseteq \{\pm 1\} = \ker K$, and $z \mapsto z^2$, that is $(x, y) \mapsto (x^2 - y^2, 2xy)$, is a nonzero element of $\mathrm{Poly}_2(L,K)^G$. Nevertheless every equivariant piecewise linear map $F : L \to K$ is zero. Indeed, G is connected and $L^G = 0$, so by the structure theorem of [Aie26] recalled in Section 1.3, F is an equivariant linear map $L \to K$; since $L \not\cong K$, it vanishes by Schur's lemma. The interpolation argument in Theorem 4.2.3 requires a finite free orbit and therefore does not apply to this $SO(2)$ action.

For irreducible L, K, combining Theorem 4.2.3 with Theorem 2.6.1 identifies the GTW existence criterion with $D(L,K) \neq \emptyset$. The following refinement also decides whether each parity occurs.

Corollary 4.2.5. *Let $L \neq 0$ and let K be irreducible with $\ker L \subseteq \ker K$. Put $Q = G/\ker L$, so that Q acts faithfully on L and naturally on K.*

(a) *If no element of Q acts as $-I_L$, then both even and odd degrees occur, with*
$$d_0(L,K),\ d_1(L,K) \le 2|Q| - 1.$$

(b) *If some $z \in Q$ acts as $-I_L$, then z is a central involution and acts on K as either I_K or $-I_K$. In the first case only even degrees occur, and in the second only odd degrees occur. In either case, the occurring parity has threshold at most $|Q| - 1$.*

Proof. Fix $\varepsilon \in \{0,1\}$ and let $\widetilde{Q} = Q \times \{\pm 1\}$ act on L and K by
$$(q,t)\cdot x = t\,qx, \qquad (q,t)\cdot y = t^{\varepsilon} qy.$$
A nonzero homogeneous Q-equivariant polynomial $P : L \to K$ is $\widetilde{Q}$-equivariant exactly when its degree is congruent to ε modulo 2.

Suppose first that no element of Q acts as $-I_L$. Then the $\widetilde{Q}$-action on L is faithful. By Theorem 4.2.3, there exists a nonzero $\widetilde{Q}$-equivariant homogeneous polynomial of degree at most $|\widetilde{Q}| - 1 = 2|Q| - 1$. Applying this for $\varepsilon = 0$ and $\varepsilon = 1$ gives both parities.

Now suppose $z \in Q$ acts as $-I_L$. Since Q acts faithfully on L, we have $z^2 = 1$. Moreover, for every $q \in Q$, both qzq^{-1} and z act as $-I_L$. By faithfulness, $qzq^{-1} = z$, so z is central. The kernel of the $\widetilde{Q}$-action on L is therefore $\{(1,1),(z,-1)\}$.

By Theorem 4.2.3, a degree of parity ε occurs exactly when this kernel acts trivially on K, that is, when
$$(-1)^{\varepsilon} zy = y \qquad \text{for all } y \in K.$$

Since z is central, its action on K lies in $\mathrm{End}_G(K)$. Moreover $z^2 = 1$, so irreducibility implies that z acts as either I_K or $-I_K$. Hence exactly one parity occurs: even if z acts as I_K, and odd if it acts as $-I_K$. The faithful quotient has order $|Q|$, so the corresponding threshold is at most $|Q| - 1$. □

4.3. Computing degree sets

The detection theorem decides whether some degree occurs, but not where the first occurrences lie. Molien's formula packages the dimensions in all degrees into one generating series.

Proposition 4.3.1 (Molien series). *Let G be finite and $c_k = \dim \mathrm{Hom}_G(\mathrm{Sym}^k L, K)$. Then*
$$\sum_{k\ge 0} c_k t^k = \frac{1}{|G|}\sum_{g\in G} \frac{\chi_K(g)}{\det(I - t\rho_L(g))} = \frac{1}{|G|}\sum_{g\in G} \chi_K(g)\exp\Big(\sum_{m\ge 1}\frac{t^m}{m}\chi_L(g^m)\Big).$$
Here $\rho_L : G \to \mathrm{GL}(L)$ is the homomorphism defining the action of G on L, and χ_L and χ_K are the characters of L and K.

Proof. Diagonalising $\rho_L(g)$ over $\mathbb{C}$ gives $\sum_k \chi_{\mathrm{Sym}^k L}(g) t^k = \det(I - t\rho_L(g))^{-1}$, whose logarithm is $\sum_{m\ge 1}\frac{t^m}{m}\operatorname{tr}\rho_L(g^m) = \sum_{m\ge 1}\frac{t^m}{m}\chi_L(g^m)$. Real characters are self-conjugate, so $c_k = \langle \chi_{\mathrm{Sym}^k L}, \chi_K\rangle$, and averaging over G gives both forms. □

The exponential form is useful computationally because it uses only the power-map characters $\chi_L(g^m)$, which are fixed-point counts for permutation representations. We use this form for subset representations in Remark B2.1.

For an irreducible target K, the coefficient c_k records the multiplicity weighted by the real dimension of its endomorphism algebra:

$$c_k = [\mathrm{Sym}^k L : K]\ \dim_{\mathbb{R}} \mathrm{End}_G(K).$$

Example 4.3.2 (The standard representation of S_3). Let L_1 be the standard representation of S_3. On the classes of e, (12) and (123), of sizes 1, 3 and 2, we have $\det(I - t\rho_{L_1}(g)) = (1-t)^2$, $1 - t^2$ and $1 + t + t^2$, respectively. Substituting these and the character table of Example 2.3.3 into Proposition 4.3.1 gives

$$\sum_k c_k(\mathbf{1})t^k = \frac{1}{(1-t^2)(1-t^3)}, \qquad \sum_k c_k(\varepsilon)t^k = \frac{t^3}{(1-t^2)(1-t^3)},$$
$$\sum_k c_k(L_1)t^k = \frac{t+t^2}{(1-t^2)(1-t^3)}.$$

The first series is the Hilbert series of the invariant ring of the reflection group S_3, generated by the elementary symmetric polynomials of degrees 2 and 3 restricted to L_1, and its coefficient is nonzero in every degree except 1. The resulting degree sets are recorded in Figure 4.1, together with those of the one-dimensional sources from Example 4.3.3 below. For the targets $\mathbf{1}$, ε and L_1, the thresholds of Proposition 4.2.2 are $(d_0, d_1) = (0, 3)$, $(6, 3)$ and $(2, 1)$. Both parities occur in each case, as predicted by Corollary 4.2.5, since no element of S_3 acts as $-I$ on L_1.

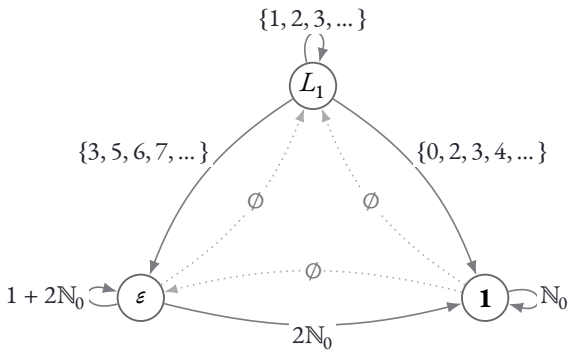


Figure 4.1. The irreducible representations of S_3, with each arrow $L \to K$ labelled by the degree set $D(L, K)$ of Example 4.3.2. Dotted arrows violate the kernel criterion and have empty degree set.

In small representations one can often read the degree set directly from the symmetric powers. The next two examples illustrate Theorem 4.2.3 and Proposition 4.2.2.

Example 4.3.3 (One-dimensional sources). Suppose $\dim L = 1$. Then $\mathrm{Sym}^k L = L^{\otimes k}$. If χ is the real character of L, its order is $q \in \{1, 2\}$. For the one-dimensional target χ^a, $a \in \{0, 1\}$,

$$D(L, \chi^a) = \{k \geq 0 : k \equiv a \pmod{q}\}.$$

Thus a trivial source reaches the trivial target in every degree, while a source of order two reaches the trivial target in even degrees and itself in odd degrees.

Example 4.3.4 (Rotation representations of C_n). Let $C_n = \langle a \rangle$, and let ρ_s and ρ_t be irreducible two-dimensional real rotation representations, with $2s, 2t \not\equiv 0 \pmod{n}$. Writing $(\rho_s)_{\mathbb{C}} \cong W_s \oplus W_{-s}$, define

$$R_k = \{r \in \mathbb{Z} : |r| \le k,\ r \equiv k \pmod 2\}.$$

Then $\mathrm{Sym}^k((\rho_s)_{\mathbb{C}}) \cong \bigoplus_{r \in R_k} W_{rs}$, and therefore

$$D(\rho_s, \rho_t) = \{k \ge 0 : \exists r \in R_k \text{ with } rs \equiv \pm t \pmod n\}.$$

This set is nonempty exactly when $\gcd(s, n) \mid t$, or equivalently when $\ker \rho_s \subseteq \ker \rho_t$. Since $R_k \subseteq R_{k+2}$, occurrence persists within each parity, as predicted by Proposition 4.2.2.

Proof of the claims in Example 4.3.4. By Example 2.3.2, $(\rho_s)_{\mathbb{C}} = W_s \oplus W_{-s}$, where a acts on a vector $u \in W_s$ by ζ^s and on a vector $v \in W_{-s}$ by ζ^{-s}. The monomials $u^p v^q$ with $p + q = k$ form a basis of $\mathrm{Sym}^k((\rho_s)_{\mathbb{C}})$, and a acts on $u^p v^q$ by $\zeta^{(p-q)s}$. As (p, q) ranges over pairs with $p + q = k$, the integer $r = p - q$ ranges over R_k. This proves the decomposition of $\mathrm{Sym}^k((\rho_s)_{\mathbb{C}})$. By the complexification criterion following (4), $\mathrm{Hom}_{C_n}(\mathrm{Sym}^k \rho_s, \rho_t) \neq 0$ if and only if $\mathrm{Sym}^k((\rho_s)_{\mathbb{C}})$ and $W_t \oplus W_{-t}$ share a constituent, that is, if and only if $rs \equiv \pm t \pmod n$ for some $r \in R_k$.

Every integer r lies in R_k for $k = |r|$, so $D(\rho_s, \rho_t) \neq \emptyset$ if and only if t or $-t$ lies in the subgroup $s\mathbb{Z} + n\mathbb{Z} = \gcd(s, n)\mathbb{Z}$ of $\mathbb{Z}$, that is, if and only if $\gcd(s, n) \mid t$. Finally $\ker \rho_s$ is the subgroup of order $\gcd(s, n)$, namely $\langle a^{n/\gcd(s,n)} \rangle$, and it acts trivially on ρ_t exactly when $\frac{n}{\gcd(s,n)} t \equiv 0 \pmod n$, that is, when $\gcd(s, n) \mid t$. □

The same calculation handles one-dimensional targets: $\mathrm{Hom}_{C_n}(\mathrm{Sym}^k \rho_s, W_t) \neq 0$ for $t \in \{0, n/2\}$ if and only if $rs \equiv t \pmod n$ for some $r \in R_k$. For example, $d(\rho_s, \mathbf{1}) = 0$, attained by $r = 0$. We use this in Section 6.3.

Chapter 5

Coordinatewise nonlinearities

The preceding chapters determine which equivariant polynomial maps exist. We now ask which of these maps a coordinatewise layer produces.

The input representation L is arbitrary throughout and need not be irreducible or multiplicity-free. In particular, the results include interactions between components of a direct-sum input, without requiring a separate treatment of those components.

5.1. Coordinatewise layers and their degree components

There are two pieces of data to track: the polynomial produced by the hidden coordinates, and the scalar Hermite coefficient supplied by the activation. We introduce them separately.

Let L and K be finite-dimensional real orthogonal G-representations, regarded as the input and output spaces. We ensure equivariance of coordinatewise application by choosing a basis that the group permutes. A *hidden representation* is an intermediate representation $H = \mathbb{R}^{\mathscr{B}}$ with an orthonormal basis $\mathscr{B}$ permuted by G. Throughout this chapter we assume that the action on $\mathscr{B}$ is transitive.

No inclusion relations between L, H and K are assumed. They are connected by equivariant linear maps

$$A : L \longrightarrow H \quad \text{(lift)}, \qquad R : H \longrightarrow K \quad \text{(readout)}.$$

Thus a coordinatewise layer has the form

$$L \xrightarrow{A} H \xrightarrow{\sigma_{\mathscr{B}}} H \xrightarrow{R} K,$$

where the nonlinearity is applied in the distinguished coordinate basis of H.

Hidden representations with several coordinate orbits are treated in Remark 5.1.8.

Definition 5.1.1. For an activation $\sigma : \mathbb{R} \to \mathbb{R}$, its *coordinatewise extension* to H is

$$\sigma_{\mathscr{B}}\left(\sum_{b\in\mathscr{B}} z_b b\right) = \sum_{b\in\mathscr{B}} \sigma(z_b) b.$$

The associated *coordinatewise layer* is

$$F : L \longrightarrow K, \qquad F(x) = R\sigma_{\mathscr{B}}(Ax).$$

By Lemma 2.3.1, the layer F (Figure 5.1) is equivariant. We assume $A \neq 0$.

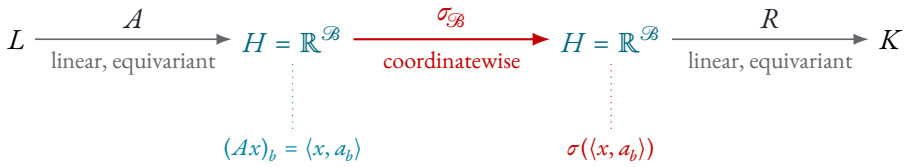


Figure 5.1. A coordinatewise layer $F = R \circ \sigma_{\mathscr{B}} \circ A$. The group G acts orthogonally on L and K and permutes the basis $\mathscr{B}$ of the hidden representation H. The degree-k component of F is determined by the Gram matrix $C = (\langle a_b, a_{b'} \rangle)$ of the coordinate vectors $a_b = A^* b$, the readout R and the scalar Hermite coefficient $a_k(\sigma; s)$.

Definition 5.1.2. For $b \in \mathscr{B}$, define the coordinate vector $a_b = A^* b \in L$. Thus the bth hidden coordinate is $(Ax)_b = \langle x, a_b \rangle$. The *coordinate Gram matrix* is

$$C = (C_{bb'})_{b,b' \in \mathscr{B}}, \qquad C_{bb'} = \langle a_b, a_{b'} \rangle.$$

Thus C is the matrix of AA^* in the basis $\mathscr{B}$.

Lemma 5.1.3. *There is an $s > 0$ such that $\|a_b\| = s$ for every $b \in \mathscr{B}$. Moreover,*

$$s^2 = C_{bb} = \frac{\operatorname{tr} C}{\dim H}.$$

For $X \sim \gamma_L$, every coordinate $(AX)_b$ therefore has distribution $N(0, s^2)$.

Proof. Equivariance and orthogonality give $a_{gb} = g a_b$. Transitivity implies that all coordinate vectors have the same norm, and $A \neq 0$ implies that this norm is positive. Summing the diagonal entries of C gives the trace identity. Finally, $\langle X, a_b \rangle$ is centred Gaussian with variance $\|a_b\|^2$. □

The activation enters the kth chaos through a single scalar coefficient.

Definition 5.1.4. Suppose $\sigma(sZ)$ is square-integrable for $Z \sim N(0,1)$. Its *Hermite coefficients at scale s* are

$$a_k(\sigma; s) = \frac{1}{k!} \mathbb{E}[\sigma(sZ) \mathrm{He}_k(Z)], \qquad a_k(\sigma) := a_k(\sigma; 1).$$

We impose this integrability assumption throughout. It holds for every continuous piecewise linear activation with finitely many knots, since such a function has at most linear growth.

The remaining degree-k information comes from the fixed lift and readout. It is encoded by the coordinatewise kth power.

Definition 5.1.5. For $z = \sum_b z_b b \in H$, define

$$z^{\odot k} := \sum_b z_b^k b \quad (k \geq 0).$$

The *degree-k coordinate polynomial* associated with A and R is

$$Q_k : L \longrightarrow K, \qquad Q_k(x) = R\left((Ax)^{\odot k}\right) = \sum_{b \in \mathscr{B}} \langle x, a_b \rangle^k R b.$$

For matrices, $\circ$ denotes the entrywise (Hadamard, or Schur) product, and $C^{\circ k}$ denotes the entrywise power, with $C^{\circ 0} = J$, the all-ones matrix.

Coordinatewise powers commute with permutations, so $Q_k \in \mathrm{Poly}_k(L, K)^G$. By Lemma 3.4.2, on homogeneous degree-k polynomials, Π_k is an isomorphism onto $\mathscr{H}_k(L)$. In particular,

$$\Pi_k Q_k = 0 \quad \Longleftrightarrow \quad Q_k = 0.$$

All L^2 norms below use the standard Gaussian on the source and the fixed inner product on the target.

With these definitions, the activation and the coordinate geometry separate as follows.

Theorem 5.1.6 (Coordinatewise factorisation). *Let $H = \mathbb{R}^{\mathscr{B}}$ be a transitive permutation representation, let $A : L \to H$ and $R : H \to K$ be equivariant linear maps with $A \neq 0$, let s be the common coordinate scale of Lemma 5.1.3, let $\sigma : \mathbb{R} \to \mathbb{R}$ be such that $\sigma(sZ)$ is square-integrable for $Z \sim N(0,1)$, and let $F(x) = R\sigma_{\mathscr{B}}(Ax)$. Then, for every $k \geq 0$,*

$$\Pi_k F = \frac{a_k(\sigma; s)}{s^k}\, \Pi_k Q_k, \tag{8}$$

$$\|\Pi_k F\|_{L^2}^2 = \frac{k!\, a_k(\sigma; s)^2}{s^{2k}}\, \mathrm{tr}\left(RC^{\circ k}R^*\right).$$

Consequently,

$$\Pi_k F \neq 0 \quad \Longleftrightarrow \quad a_k(\sigma; s) \neq 0 \text{ and } Q_k \neq 0,$$
$$Q_k \neq 0 \quad \Longleftrightarrow \quad \mathrm{tr}(RC^{\circ k}R^*) > 0.$$

No irreducibility or multiplicity-free assumption is required.

The transitivity assumption is used only to give all hidden coordinates the same variance; see Remark 5.1.8 for several coordinate orbits.

Proof. For $b \in \mathscr{B}$, write $\ell_b(x) = \langle x, a_b \rangle$. Since $\ell_b(X)/s \sim N(0,1)$ for $X \sim \gamma_L$, pulling back the Hermite expansion in Proposition 3.3.2 gives

$$\sigma(\ell_b) = \sum_{j \geq 0} a_j(\sigma; s)\mathrm{He}_j(\ell_b/s) \quad \text{in } L^2(\gamma_L).$$

By Lemma 3.4.3, $\mathrm{He}_k(\ell_b/s) = s^{-k}\Pi_k(\ell_b^k)$. Hence, by linearity,

$$\Pi_k F = \sum_{b \in \mathscr{B}} \Pi_k[\sigma(\ell_b)]\, Rb = \frac{a_k(\sigma; s)}{s^k}\Pi_k Q_k,$$

which proves (8).

By Lemma 3.4.4, $\left\langle \Pi_k(\ell_b^k), \Pi_k(\ell_{b'}^k) \right\rangle_{L^2} = k!\, C_{bb'}^k$. Therefore

$$\|\Pi_k Q_k\|_{L^2}^2 = k! \sum_{b,b' \in \mathscr{B}} C_{bb'}^k \left\langle Rb, Rb' \right\rangle = k!\, \mathrm{tr}\left(C^{\circ k}R^*R\right) = k!\, \mathrm{tr}\left(RC^{\circ k}R^*\right).$$

This also holds for $k = 0$, with $C^{\circ 0} = J$. Combining it with (8) gives the stated norm formula.

Finally, since Q_k is homogeneous of degree k, Lemma 3.4.2 gives $\Pi_k Q_k = 0$ if and only if $Q_k = 0$. Thus the Gram trace is nonnegative and vanishes exactly when $Q_k = 0$, while (8) yields

$$\Pi_k F \neq 0 \iff a_k(\sigma; s) \neq 0 \text{ and } Q_k \neq 0.$$

□

The theorem separates the three questions *(a)–(c)* of the introduction. For fixed A and R, changing the activation only changes the scalar multiplying $\Pi_k Q_k$; it cannot select a different polynomial direction at degree k.

Example 5.1.7. Let $H = \mathbb{R}^3$ be the S_3 permutation representation, and recall the trivial and standard subrepresentations, L_0 and L_1.

Let $A : L_1 \hookrightarrow H$ be inclusion, and $R = p_{L_0}$ be orthogonal projection. Then $C = p_{L_1}$ and $s^2 = \dim L_1/3 = 2/3$. Let $\mathbf{1} = (1, 1, 1)$. Then from (2),

$$Q_2(x) = \frac{1}{3}\|x\|^2 \mathbf{1}, \qquad \Pi_2 Q_2(x) = \frac{1}{3}(\|x\|^2 - 2)\mathbf{1}.$$

Here $\mathbb{E}[\|X\|^2] = 2$ for $X \sim \gamma_{L_1}$. Thus

$$\Pi_2 F(x) = \frac{a_2(\sigma; \sqrt{2/3})}{2}(\|x\|^2 - 2)\mathbf{1}.$$

The coordinate polynomial is nonzero, so this second-chaos interaction is present precisely when $a_2(\sigma; \sqrt{2/3}) \neq 0$.

Remark 5.1.8 (Several coordinate orbits). Transitivity is used in Theorem 5.1.6 only to give all hidden coordinates the same variance. For an arbitrary permutation basis $\mathcal{B}$, put $C = AA^*$ and $s_b = \sqrt{C_{bb}}$, which is constant on coordinate orbits, and let D_k be the diagonal matrix with entries $a_k(\sigma; s_b)/s_b^k$ when $s_b > 0$, and $\sigma(0)\delta_{k0}$ when $s_b = 0$. Then D_k is equivariant, and expanding each coordinate at its own scale exactly as in the proof of Theorem 5.1.6 gives

$$\Pi_k F = \Pi_k \left[R D_k \left((Ax)^{\odot k} \right) \right], \qquad \|\Pi_k F\|_{L^2}^2 = k!\ \mathrm{tr}\left(R D_k C^{\circ k} D_k R^* \right).$$

Contributions of different orbits can cancel after the readout, so separate nonzero tests for the orbits do not replace this combined formula.

5.2. The submodule generated by the coordinates

We next explain why the polynomial Q_k can vanish even when $\mathrm{Hom}_G(\mathrm{Sym}^k L, K)$ is nonzero.

Definition 5.2.1. For $k \geq 0$, define

$$Y_k = \mathrm{span}\{a_b^k : b \in \mathcal{B}\} \subseteq \mathrm{Sym}^k L.$$

We call Y_k the *coordinate-generated submodule* of degree k. Define also

$$\phi_k : H \longrightarrow \mathrm{Sym}^k L, \qquad \phi_k(b) = a_b^k.$$

The relation $a_{gb} = g a_b$ implies that ϕ_k is equivariant. Thus $Y_k = \mathrm{im}\, \phi_k$ is indeed a submodule.

Proposition 5.2.2. *For every $k \geq 0$, $\phi_k^* \phi_k = C^{\circ k}$, and so*

$$Y_k \cong (\ker C^{\circ k})^\perp$$

as G-representations. Moreover $Q_k(x) = R\phi_k^*(x^k)$, so

$$Q_k \neq 0 \quad \Longleftrightarrow \quad R\phi_k^* \neq 0.$$

Proof. The tensor inner product gives

$$\langle a_b^k, a_{b'}^k \rangle = \langle a_b, a_{b'} \rangle^k = C_{bb'}^k.$$

Hence $C^{\circ k}$ is the Gram matrix of the vectors a_b^k, proving $\phi_k^* \phi_k = C^{\circ k}$. In particular, $\ker \phi_k = \ker C^{\circ k}$. Restricting ϕ_k to the orthogonal complement of its kernel gives the stated isomorphism onto Y_k.

For $z \in \mathrm{Sym}^k L$,

$$R\phi_k^* z = \sum_b \langle z, a_b^k \rangle Rb.$$

Taking $z = x^k$ gives $Q_k(x)$, in accordance with Proposition 2.4.3. Since pure powers span $\mathrm{Sym}^k L$, the polynomial vanishes identically if and only if $R\phi_k^*$ is zero. □

An arbitrary degree-k equivariant polynomial may use any direction in $\mathrm{Sym}^k L$, whereas the coordinate construction factors through the coordinate-generated submodule Y_k. A target can therefore occur in $\mathrm{Sym}^k L$ and still be invisible to the chosen coordinates. Even when it occurs in Y_k, the readout R may kill the resulting direction.

5.3. Projection onto irreducible summands

When the lift is an inclusion and the readout is an orthogonal projection, the Gram criterion becomes especially transparent. Let $W, K \subseteq H$ be nonzero subrepresentations and take

$$A = \iota : W \hookrightarrow H, \qquad R = p_K : H \longrightarrow K.$$

Writing E_W and E_K for the orthogonal projector matrices in $\mathcal{B}$, we have

$$C = E_W, \qquad s^2 = \frac{\dim W}{\dim H}, \qquad Q_k(x) = p_K(x^{\odot k}) \quad (x \in W).$$

The norm formula becomes

$$\|\Pi_k Q_k\|^2_{L^2(\gamma_W;K)} = k!\ \mathrm{tr}(E_K E_W^{\circ k}),$$

which holds without a multiplicity-free assumption.

Suppose now that H is multiplicity-free, meaning that $H = \bigoplus_{j=0}^m L_j$, where the L_j are pairwise non-isomorphic irreducible representations. Let L_0 be the trivial summand, let E_j be orthogonal projection onto L_j, and let $d_j = \dim L_j$. Transitivity gives

$$E_0 = \frac{J}{\dim H}, \qquad (E_j)_{bb} = \frac{d_j}{\dim H}.$$

Multiplicity-freeness and self-adjointness make $E_W^{\circ k}$ scalar on each irreducible summand. We record those scalars next.

Definition 5.3.1. For the fixed source subrepresentation W (not necessarily irreducible), define

$$\kappa_j(k) := \frac{1}{d_j}\operatorname{tr}(E_j E_W^{\circ k}), \qquad k \geq 0.$$

We call $\kappa_j(k)$ the *projector coefficient* for the source W, target L_j and degree k.

Corollary 5.3.2. *For each $k \geq 0$:*

(a) *The Hadamard power of the projector decomposes as*

$$E_W^{\circ k} = \sum_{j=0}^{m} \kappa_j(k) E_j, \qquad \kappa_j(k) \geq 0.$$

(b) *For $Q_k^{(j)}(x) = p_j(x^{\odot k})$ with $x \in W$,*

$$Q_k^{(j)} \neq 0 \iff \kappa_j(k) > 0, \qquad \left\| \Pi_k Q_k^{(j)} \right\|_{L^2}^2 = k!\, d_j\, \kappa_j(k).$$

(c) *The coordinate-generated submodule for the inclusion $W \hookrightarrow H$ is*

$$Y_k \cong \bigoplus_{j:\kappa_j(k)>0} L_j.$$

Proof. Hadamard products of G-equivariant matrices are again equivariant. Thus $E_W^{\circ k}$ is a self-adjoint G-endomorphism of $H = \bigoplus_{j=0}^m L_j$. By multiplicity-freeness and Schur's lemma, together with self-adjointness, it acts on each L_j by a real scalar. Hence $E_W^{\circ k} = \sum_{j=0}^m \lambda_j(k) E_j$. Multiplying by E_j and taking traces gives $\lambda_j(k) = \kappa_j(k)$, so $E_W^{\circ k} = \sum_{j=0}^m \kappa_j(k) E_j$.

Since E_W is positive semidefinite, the Schur product theorem [HJ13] implies that $E_W^{\circ k}$ is positive semidefinite. Its scalar on L_j is therefore nonnegative, so $\kappa_j(k) \geq 0$.

For the inclusion $\iota : W \hookrightarrow H$ and projection $p_j : H \to L_j$, the norm formula gives

$$\left\| \Pi_k Q_k^{(j)} \right\|_{L^2}^2 = k!\, \operatorname{tr}(E_j E_W^{\circ k}) = k!\, d_j \kappa_j(k).$$

By Lemma 3.4.2, Π_k is injective on homogeneous degree-k polynomials, so

$$Q_k^{(j)} \neq 0 \iff \kappa_j(k) > 0.$$

Finally, Proposition 5.2.2 gives $Y_k \cong (\ker E_W^{\circ k})^\perp$. Since $E_W^{\circ k}$ acts on L_j by $\kappa_j(k)$, its kernel is the sum of those L_j with $\kappa_j(k) = 0$. Hence

$$Y_k \cong \bigoplus_{j:\kappa_j(k)>0} L_j.$$

□

When W is an irreducible summand and $k = 2$, these Hadamard coefficients are the usual Krein parameters [BCN89], the structure constants for entrywise products of primitive idempotents, up to the chosen normalisation. Higher k gives iterated Hadamard products. For subset modules these are the Krein parameters of the Johnson scheme; see Section 8.2.

Lemma 5.3.3. *For each $k \geq 0$,*

$$\kappa_j(k) > 0 \implies \kappa_j(k+2) > 0.$$

Equivalently, if $p_j(x^{\odot k})|_W$ is nonzero, then $p_j(x^{\odot(k+2)})|_W$ is nonzero. Hence, for each j, the supported degrees form at most one even tail and one odd tail.

Proof. For the trivial summand L_0,

$$\kappa_0(2) = \operatorname{tr}(E_0 E_W^{\circ 2}) = \frac{\dim W}{\dim H}.$$

Using the decomposition from Corollary 5.3.2,

$$E_W^{\circ 2} = \frac{\dim W}{\dim H} E_0 + B = \frac{\dim W}{(\dim H)^2} J + B,$$

where B is positive semidefinite. Therefore

$$E_W^{\circ(k+2)} = E_W^{\circ k} \circ E_W^{\circ 2} = \frac{\dim W}{(\dim H)^2} E_W^{\circ k} + E_W^{\circ k} \circ B.$$

The second term is positive semidefinite by the Schur product theorem. Its projector coefficients are therefore nonnegative, so

$$\kappa_j(k+2) \geq \frac{\dim W}{(\dim H)^2} \kappa_j(k).$$

Since $W \neq 0$, positivity of $\kappa_j(k)$ implies positivity of $\kappa_j(k+2)$. The equivalence with the coordinate polynomial follows from Corollary 5.3.2. □

This is the coordinate-level analogue of Proposition 4.2.2: within each parity, coordinate support also persists once it appears.

5.4. Examples of realisation and obstruction

The first example shows that a permitted quadratic map can be missed by the coordinate construction.

Example 5.4.1. Let S_4 act on the six unordered pairs of $[4] := \{1, 2, 3, 4\}$, with permutation representation $H = \mathbb{R}^{\binom{[4]}{2}}$. Its standard subrepresentation is

$$W = \left\{ x_{ab} = u_a + u_b : \sum_a u_a = 0 \right\} \cong S^{(3,1)}.$$

Indeed, the equivariant map $u \mapsto x$ has inverse $u_a = \frac{1}{2} \sum_{b \neq a} x_{ab}$ on its image.

Let $J_c e_A = e_{A^c}$ be the complementation operator. For $x \in W$, we have $x_{A^c} = -x_A$, so W is the (-1)-eigenspace of J_c. Every even coordinatewise power $x^{\odot 2m}$ lies in its $(+1)$-eigenspace, which is orthogonal to W. Hence

$$p_W(x^{\odot 2m}) = 0 \qquad (x \in W,\ m \geq 1).$$

Nevertheless,

$$u \longmapsto \left(u_a^2 - \frac{1}{4} \sum_b u_b^2 \right)_{a=1}^4$$

is a nonzero equivariant quadratic self-map of the standard vertex representation. Transporting it to W gives $\mathrm{Hom}_{S_4}(\mathrm{Sym}^2 W, W) \neq 0$. Thus symmetry permits a quadratic self-interaction, but the fixed layer $F(x) = p_W \sigma_{\mathcal{B}}(x)$ has no second chaos component. This holds for every activation satisfying the integrability assumption, including biased ReLU, because its coordinate polynomial Q_2 is zero. The same argument applies in every even degree.

We now identify the coordinate-generated submodule Y_2. Since W is the (-1)-eigenspace of J_c, its orthogonal projector is

$$E_W = \frac{1}{2}(I - J_c), \qquad E_W^{\circ 2} = \frac{1}{4}(I + J_c).$$

In particular, $\mathrm{tr}(E_W E_W^{\circ 2}) = 0$. Using $H \cong S^{(4)} \oplus S^{(3,1)} \oplus S^{(2,2)}$ and the Gram description of Y_2, we obtain

$$Y_2 \cong S^{(4)} \oplus S^{(2,2)}.$$

The coordinate-generated submodule therefore accounts for three of the six dimensions of $\mathrm{Sym}^2 W$. The quadratic map above supplies a constituent $S^{(3,1)}$, so dimension counting gives

$$\mathrm{Sym}^2 W \cong S^{(4)} \oplus S^{(3,1)} \oplus S^{(2,2)}.$$

The coordinates therefore capture every constituent except the one needed for the quadratic self-interaction of W.

For regular representations of finite abelian groups, every permitted degree has a nonzero coordinate polynomial.

Proposition 5.4.2. *Let G be finite abelian, and let $H = \mathbb{R}[G] = \bigoplus_j L_j$ be its real regular representation, equipped with the regular coordinate basis. Let $W \subseteq H$ be any nonzero subrepresentation, and let p_j be orthogonal projection onto L_j. Then, for every $k \geq 0$,*

$$p_j(x^{\odot k})|_W \not\equiv 0 \quad \Longleftrightarrow \quad \mathrm{Hom}_G(\mathrm{Sym}^k W, L_j) \neq 0.$$

Proof. After complexification, identify $H_{\mathbb{C}}$ with functions $G \to \mathbb{C}$. Since G is abelian, its regular representation is a direct sum of one-dimensional character spaces. The subrepresentation $W_{\mathbb{C}}$ is therefore spanned by a subset of the character functions.

Consider the equivariant multiplication map

$$\mu_k : \mathrm{Sym}^k W_{\mathbb{C}} \longrightarrow H_{\mathbb{C}}, \qquad f_1 \cdots f_k \longmapsto f_1 \odot \cdots \odot f_k.$$

A monomial in character functions is sent to their pointwise product, which is a nonzero character function of the same representation type. Thus every character occurring in $\mathrm{Sym}^k W_{\mathbb{C}}$ occurs in $\mathrm{im}\, \mu_k$, and conversely.

The polynomial $p_j(x^{\odot k})$ corresponds to $p_j \mu_k$. Since pure powers span the symmetric power, this polynomial is nonzero exactly when a character of $(L_j)_{\mathbb{C}}$ occurs in $\mathrm{Sym}^k W_{\mathbb{C}}$. That is, $\mathrm{Hom}_G(\mathrm{Sym}^k W, L_j) \neq 0$. □

The proposition detects nonzero degrees only; it does not say that the selected polynomial spans the full equivariant polynomial space. For these regular abelian representations, the presence of a permitted degree in the layer is therefore decided by the activation coefficient in Theorem 5.1.6.

5.5. Graded interaction graphs

We now return to the interaction graphs of Section 2.6 and explain how the results of this chapter refine them.

Let $H = \mathbb{R}^{\mathcal{B}}$ be a transitive permutation representation with an orthogonal decomposition $H = \bigoplus_{i=1}^{r} L_i$ into irreducible summands, with inclusions ι_i and orthogonal projections p_i as before, and write E_i for the matrix of $\iota_i p_i$ in the basis $\mathcal{B}$ and $d_i = \dim L_i$. Let σ be a continuous activation for which $\sigma(sZ)$ is square-integrable for all $s > 0$.

The block $(\sigma_{\mathcal{B}})_i^j = p_j \circ \sigma_{\mathcal{B}} \circ \iota_i$ of the coordinatewise activation is exactly the coordinatewise layer of Theorem 5.1.6. Its coordinate Gram matrix is $C = E_i$, its scale is $s_i = \sqrt{d_i / \dim H}$, and its coordinate polynomials are $Q_k^{ij}(x) = p_j(x^{\odot k})$ for $x \in L_i$.

Definition 5.5.1. For $k \geq 0$, define three directed graphs on the vertices $L_1, \dots, L_r$:

(a) the *degree-k symmetry graph* $\Gamma_{\mathrm{sym}}^{(k)}$, with an edge $L_i \to L_j$ when $\mathrm{Hom}_G(\mathrm{Sym}^k L_i, L_j) \neq 0$;

(b) the *degree-k coordinate graph* $\Gamma_{\mathcal{B}}^{(k)}$, with an edge when $Q_k^{ij} \neq 0$;

(c) the *degree-k interaction graph* $\Gamma_\sigma^{(k)}$ of $\sigma_{\mathcal{B}}$, with an edge when $\Pi_k(\sigma_{\mathcal{B}})_i^j \neq 0$.

Proposition 5.5.2. *For every $k \geq 0$,*

$$\Gamma_\sigma^{(k)} = \left\{ L_i \to L_j \in \Gamma_{\mathcal{B}}^{(k)} : a_k(\sigma; s_i) \neq 0 \right\} \subseteq \Gamma_{\mathcal{B}}^{(k)} \subseteq \Gamma_{\mathrm{sym}}^{(k)}.$$

Moreover, the interaction graph of Definition 2.6.2 is $\Gamma_{\sigma_{\mathcal{B}}} = \bigcup_k \Gamma_\sigma^{(k)}$, and $\bigcup_k \Gamma_{\mathrm{sym}}^{(k)} = \Gamma_{\ker}$.

Proof. The equality and the first inclusion are Theorem 5.1.6 applied to each block. The second inclusion holds because $Q_k^{ij} \in \mathrm{Poly}_k(L_i, L_j)^G \cong \mathrm{Hom}_G(\mathrm{Sym}^k L_i, L_j)$. Each block is continuous and square-integrable, so by Corollary 4.1.2 it vanishes if and only if all of its chaos components vanish; this gives $\Gamma_{\sigma_{\mathcal{B}}} = \bigcup_k \Gamma_\sigma^{(k)}$. The last statement is Theorem 4.2.3 for irreducible L_i, L_j. □

The three graphs, and the two inclusions between them, correspond to the three obstructions of the introduction: an edge of $\Gamma_{\ker}$ may be absent from $\Gamma_{\mathrm{sym}}^{(k)}$ in a given degree (symmetry), from $\Gamma_{\mathcal{B}}^{(k)}$ (coordinates), or from $\Gamma_\sigma^{(k)}$ (activation). By Example 5.4.1, the second inclusion can be strict: for S_4 acting on pairs, the self-loop at $S^{(3,1)}$ lies in $\Gamma_{\mathrm{sym}}^{(2)}$ but not in $\Gamma_{\mathcal{B}}^{(2)}$. On the other hand, by Proposition 5.4.2, the second inclusion is an equality for regular representations of finite abelian groups, so for these, the realised graph of an activation is determined by the degree sets and the Hermite spectrum alone. We discuss this for C_8 in Section 6.3.

Chapter 6

Activation spectra and degreewise spanning

The coordinatewise factorisation separates the polynomial selected by a layer from the coefficient supplied by its activation. We first determine when $a_k(\sigma; s)$ vanishes. We then describe what a collection of coordinatewise components can span in each degree, and give explicit spanning constructions (Sections 6.4 and 6.5).

Definition 6.0.1. Let $s > 0$, and suppose $\sigma(sZ)$ is square-integrable for $Z \sim N(0,1)$. The *Hermite spectrum of σ at scale s* is the sequence $(a_k(\sigma; s))_{k\geq 0}$. Its *support* is

$$\operatorname{supp} a_\bullet(\sigma; s) = \{k \in \mathbb{N}_0 : a_k(\sigma; s) \neq 0\}.$$

At scale one, write $\operatorname{supp} a_\bullet(\sigma) = \operatorname{supp} a_\bullet(\sigma; 1)$.

For a layer satisfying the assumptions of Theorem 5.1.6,

$$\{k : \Pi_k F \neq 0\} = \{k : Q_k \neq 0\} \cap \operatorname{supp} a_\bullet(\sigma; s).$$

Thus the activation support determines which of the nonzero coordinate polynomials contribute to the layer.

6.1. Positively homogeneous activations

We start with positively homogeneous activations, which include the identity, absolute value and ReLU.

Definition 6.1.1. A function $\sigma : \mathbb{R} \to \mathbb{R}$ is *positively homogeneous of degree one* if

$$\sigma(tx) = t\sigma(x) \qquad (t \geq 0,\ x \in \mathbb{R}).$$

Writing $m_+ = \sigma(1)$ and $m_- = -\sigma(-1)$, such a function is necessarily of the form

$$\sigma(x) = \begin{cases} m_+ x, & x \geq 0, \\ m_- x, & x \leq 0, \end{cases} \quad = \quad \frac{m_+ + m_-}{2}\, x + \frac{m_+ - m_-}{2}\, |x| \,.$$

Every such function is continuous and piecewise linear.

From here on, *homogeneous* means positively homogeneous of degree one.

Theorem 6.1.2. *Let $\sigma \neq 0$ be homogeneous, with slopes m_+ and m_- as above. For every $s > 0$,*

$$a_k(\sigma; s) = s\, a_k(\sigma),$$

and its support is given by

$$\begin{aligned} m_+ = m_- &\implies \operatorname{supp} a_\bullet(\sigma) = \{1\}, \\ m_+ = -m_- &\implies \operatorname{supp} a_\bullet(\sigma) = 2\mathbb{N}_0, \\ m_+ \neq \pm m_- &\implies \operatorname{supp} a_\bullet(\sigma) = \{1\} \cup 2\mathbb{N}_0. \end{aligned}$$

Proof. Positive homogeneity gives $\sigma(sZ) = s\sigma(Z)$, which proves the scaling identity.

The odd part of σ is a multiple of $x = \mathrm{He}_1(x)$, which contributes only in degree one. The function $|x|$ is even, so its odd Hermite coefficients vanish. Its degree-zero coefficient is $\mathbb{E}[|Z|] = 2\varphi(0) > 0$. For $r \geq 1$, two integrations by parts give

$$a_{2r}(|\cdot|) = \frac{2}{(2r)!}\mathrm{He}_{2r-2}(0)\varphi(0).$$

These coefficients are nonzero because $\mathrm{He}_{2r-2}(0) \neq 0$ by Lemma 3.3.1(d). Thus $|x|$ has support exactly $2\mathbb{N}_0$. The two parts of σ have disjoint supports, so their nonzero coefficients cannot cancel. □

Example 6.1.3. The identity activation has support $\{1\}$, absolute value has support $2\mathbb{N}_0$, and $\mathrm{ReLU}(x) := \max(x, 0) = \frac{x+|x|}{2}$ has support $\{1\} \cup 2\mathbb{N}_0$.

The scaling identity shows that changing the input scale cannot create a degree that is absent at scale one. In particular, no odd degree $k > 1$ occurs, for any σ in the class and any input scale.

Corollary 6.1.4. *Let $F(x) = R\sigma_{\mathscr{B}}(Ax)$ satisfy the assumptions of Theorem 5.1.6, with σ homogeneous. Then*

$$\Pi_k F = 0 \qquad \text{for every odd } k \geq 3.$$

The same conclusion holds for finite sums of such layers with the same source and target.

Proof. Write $\sigma(z) = \lambda z + \mu\,|z|$ with $\lambda = \frac{m_+ + m_-}{2}$ and $\mu = \frac{m_+ - m_-}{2}$. The second summand is even, so the odd part of the layer is

$$\frac{F(x) - F(-x)}{2} = \lambda\, RAx,$$

which is linear and therefore lies in the first chaos, while an even map has no odd chaos components at all. Hence $\Pi_k F = 0$ for every odd $k \geq 3$, and the same argument applies to a finite sum of such layers. □

This statement applies to a single layer, or a sum of such layers, but it does not imply the same restriction for their compositions.

6.2. Bias and input scale

We now show how a shared additive bias changes this conclusion.

Definition 6.2.1. For a bias $c \in \mathbb{R}$, define the *biased ReLU*

$$\sigma_c(x) := \mathrm{ReLU}(x - c).$$

For fixed equivariant maps $A : L \to H$ and $R : H \to K$, write $F_c(x) = R(\sigma_c)_{\mathscr{B}}(Ax)$.

The same bias is applied to every hidden coordinate, so F_c remains equivariant. The identity $\sigma_c(sz) = s\,\sigma_{c/s}(z)$ gives $a_k(\sigma_c; s) = s\,a_k(\sigma_{c/s})$. Thus the support depends on the ratio c/s. The following formula determines it explicitly.

Proposition 6.2.2. *For every $s > 0$ and $k \geq 2$,*

$$a_k(\sigma_c; s) = \frac{s}{k!}\mathrm{He}_{k-2}(c/s)\varphi(c/s).$$

Moreover, $a_0(\sigma_c; s)$ and $a_1(\sigma_c; s)$ are strictly positive.

Proof. Put $u = c/s$. For $k \geq 2$,

$$a_k(\sigma_c; s) = \frac{s}{k!}\int_u^\infty (z-u)\mathrm{He}_k(z)\varphi(z)\,dz = \frac{s}{k!}\mathrm{He}_{k-2}(u)\varphi(u),$$

where the second equality follows from two integrations by parts using $(\mathrm{He}_j\varphi)' = -\mathrm{He}_{j+1}\varphi$. For $k = 0$, the defining integral has a nonnegative integrand that is positive on (u, ∞). For $k = 1$, one integration by parts gives $a_1(\sigma_c; s) = s\int_u^\infty \varphi(z)\,dz > 0$. □

Corollary 6.2.3. *Fix $A \neq 0$ and a degree bound $D \in \mathbb{N}_0$. Let s be the common coordinate scale (Lemma 5.1.3). For all but finitely many biases $c \in \mathbb{R}$,*

$$\Pi_k F_c \neq 0 \quad \Longleftrightarrow \quad Q_k \neq 0 \qquad (0 \leq k \leq D).$$

Proof. The coefficients in degrees zero and one never vanish. For $2 \leq k \leq D$, vanishing occurs precisely when c/s is a root of He_{k-2}. Each of these finitely many polynomials has finitely many roots. Outside the resulting finite set of biases, every activation coefficient through degree D is nonzero. Apply Theorem 5.1.6. □

Thus a suitable shared bias retains every nonzero coordinate polynomial through a chosen degree. It cannot recover a degree for which $Q_k = 0$, or select additional polynomial directions with A and R fixed.

Unlike the positively homogeneous case, changing the input scale at a fixed nonzero bias can change activation support, because it changes c/s. This concerns only the scalar activation coefficients, so any change to the lift must also be accounted for in the coordinate polynomials.

6.3. An activation obstruction for C_8

The ungraded ReLU obstruction in this example is already part of the classification for cyclic groups in [GTW25, Theorem 3B.4]. Here we identify its first permitted degree and explain the effect of a bias.

The real regular representation of C_8 decomposes as

$$\mathbb{R}[C_8] \cong \mathbf{1} \oplus \varepsilon \oplus \rho_1 \oplus \rho_2 \oplus \rho_3,$$

where ε is the sign character and ρ_j is the two-dimensional representation in which a generator acts by rotation through $2\pi j/8$.

The complete table of first degrees $d(L, K)$ between the summands of $\mathbb{R}[C_8]$ (sources index rows, targets index columns) follows from Examples 4.3.3 and 4.3.4 and the remark after the proof of Example 4.3.4:

$d(L,K)$	$\mathbf{1}$	ε	ρ_1	ρ_2	ρ_3
$\mathbf{1}$	0	∞	∞	∞	∞
ε	0	1	∞	∞	∞
ρ_1	0	4	1	2	3
ρ_2	0	2	∞	1	∞
ρ_3	0	4	3	2	1

Table 6.1. First degrees $d(L, K)$ between the summands of $\mathbb{R}[C_8]$; rows are sources, columns targets.

For instance, in Table 6.1, $d(\rho_1, \varepsilon) = 4$ because the least $r \equiv 4 \pmod 8$ with $|r| \le k$ is $r = 4$, while $d(\rho_2, \varepsilon) = 2$ because $2r \equiv 4 \pmod 8$ is solved by $r = 2$. The entries equal to ∞ are exactly the pairs violating the kernel condition, for example $\ker \rho_2 = \{1, a^4\} \not\subseteq \ker \rho_1 = \{1\}$. The graph of all finite entries is shown in Figure 6.1.

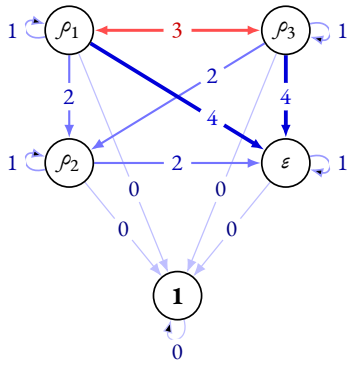


Figure 6.1. The kernel-inclusion graph $\Gamma_{\ker} = \bigcup_k \Gamma_{\mathrm{sym}}^{(k)}$ for the real irreducible summands of $\mathbb{R}[C_8]$. An arrow $L \to K$ is drawn precisely when $d(L, K) < \infty$, and is labelled and coloured by $d(L, K)$. The two red edges, between ρ_1 and ρ_3, first occur in degree three.

Consider the two projected layers from ρ_1 to ρ_3 and from ρ_3 to ρ_1. The element a^4 acts as $-I$ on ρ_1 and on ρ_3, so Corollary 4.2.5 already shows that only odd degrees occur in either direction. Example 4.3.4 locates the first of them:

$$D(\rho_1, \rho_3) = D(\rho_3, \rho_1) = \{3, 5, 7, \ldots\}.$$

Indeed, for the first direction the character condition is $r \equiv \pm 3 \pmod 8$, while for the reverse direction it is $3r \equiv \pm 1 \pmod 8$, which is equivalent to the same condition. Together with $|r| \le k$ and $r \equiv k \pmod 2$, this permits exactly the odd degrees $k \ge 3$.

By Proposition 5.4.2, the coordinate polynomials are nonzero in every permitted degree. For a positively homogeneous activation, however, Theorem 6.1.2 gives

$$a_k(\sigma; s) = 0 \qquad (k = 3, 5, 7, \ldots).$$

Hence every chaos component of either projected layer vanishes. Since these layers are continuous, vanishing almost everywhere for the Gaussian measure implies that they vanish identically.

A biased ReLU with $c \neq 0$ changes this conclusion. Its third coefficient is nonzero by Proposition 6.2.2, so both projected layers acquire a nonzero third-chaos component.

This distinguishes the example from Example 5.4.1. There, the quadratic coordinate polynomial vanishes and changing the activation cannot repair it. Here, the cubic coordinate polynomial is already nonzero and the bias supplies its missing scalar coefficient.

The complete interaction graphs

Because $\mathbb{R}[C_8]$ is the regular representation of an abelian group, Proposition 5.4.2 shows that the coordinate graph equals the symmetry graph in every degree. By Proposition 5.5.2, the interaction graph of a coordinatewise activation σ on $\mathbb{R}[C_8]$ therefore consists of those edges $L_i \to L_j$ for which $D(L_i, L_j)$ meets $\operatorname{supp} a_\bullet(\sigma; s_i)$, where $s_i^2 = \dim L_i / 8$.

Corollary 6.3.1. *Let $H = \mathbb{R}[C_8]$ with its regular basis.*

(a) *For every nonzero positively homogeneous σ with $m_+ \neq \pm m_-$, in particular for $\sigma = \mathrm{ReLU}$, the interaction graph of $\sigma_{\mathscr{B}}$ is Γ_{ker} with the two edges $\rho_1 \to \rho_3$ and $\rho_3 \to \rho_1$ removed.*

(b) *For every $c \neq 0$, the interaction graph of the biased ReLU $(\sigma_c)_{\mathscr{B}}$ is all of Γ_{ker}.*

Proof. (a) By Theorem 6.1.2, $\operatorname{supp} a_\bullet(\sigma; s) = \{1\} \cup 2\mathbb{N}_0$ for every s. Every finite entry of Table 6.1 other than $d(\rho_1, \rho_3) = d(\rho_3, \rho_1) = 3$ lies in $\{0, 1, 2, 4\}$, so all other edges are realised, while $D(\rho_1, \rho_3) = D(\rho_3, \rho_1)$ consists of odd degrees at least three and misses the support.

(b) By Proposition 6.2.2, $a_k(\sigma_c; s) \neq 0$ for $k \in \{0, 1, 2\}$, and $a_3(\sigma_c; s) = \frac{s}{6}\frac{c}{s}\varphi(c/s) \neq 0$ since $c \neq 0$. This realises every edge with first degree at most three. The remaining edges are $\rho_1 \to \varepsilon$ and $\rho_3 \to \varepsilon$, with source scale $s = \frac{1}{2}$ and $D(\rho_1, \varepsilon) = D(\rho_3, \varepsilon) = \{4, 6, 8, \ldots\}$. Now $a_4(\sigma_c; \frac{1}{2}) \propto \mathrm{He}_2(2c)$ and $a_6(\sigma_c; \frac{1}{2}) \propto \mathrm{He}_4(2c)$ cannot vanish simultaneously, since $\mathrm{He}_2(y) = 0$ forces $y^2 = 1$ and then $\mathrm{He}_4(y) = 1 - 6 + 3 = -2$. Hence these edges are realised as well. □

The ReLU graph is therefore Figure 6.1 with its two red edges removed.

6.4. The degree-k subspace spanned by several branches

The preceding results concern one branch of one layer. We now describe what a collection of such branches can span, how to control its activation coefficients, and how these statements lead to explicit polynomial layers.

With its lift, readout and activation fixed, each branch contributes at most one direction at a given degree. We first compute their span.

Fix equivariant $A_h : L \to H_h$ and $R_h : H_h \to K$ for $1 \le h \le M$, where each $H_h = \mathbb{R}^{\mathcal{B}_h}$ has a transitive coordinate basis $\mathcal{B}_h$ and $A_h \neq 0$. Write

$$Q_{h,k}(x) = R_h\left((A_h x)^{\odot k}\right), \qquad \mathcal{S}_k := \operatorname{span}\{Q_{h,k} : 1 \le h \le M\} \subseteq \operatorname{Poly}_k(L,K)^G.$$

Consider the layer $F_\theta(x) = \sum_h \theta_h R_h(\sigma_h)_{\mathcal{B}_h}(A_h x)$, with independent real output coefficients θ_h.

Proposition 6.4.1. *If $a_k(\sigma_h; s_h) \neq 0$ for all h, where $s_h^2 = \operatorname{tr}(A_h A_h^*)/\dim H_h$ (Lemma 5.1.3), then*

$$\{\Pi_k F_\theta : \theta \in \mathbb{R}^M\} = \Pi_k \mathcal{S}_k.$$

The dimension of this space is the rank of the positive semidefinite Gram matrix

$$\mathsf{G}^{(k)}_{h\ell} = k!\ \operatorname{tr}\left(R_h (A_h A_\ell^*)^{\circ k} R_\ell^*\right).$$

Here $A_h A_\ell^* : H_\ell \to H_h$ is represented in the two coordinate bases; its entrywise power is therefore well defined even when H_h and H_ℓ have different dimensions. In particular, the whole degree-k equivariant space is spanned exactly when this rank equals $\dim \operatorname{Hom}_G(\operatorname{Sym}^k L, K)$.

Proof. Fix the degree k. By Theorem 5.1.6, the hth branch contributes

$$\Pi_k\left(R_h(\sigma_h)_{\mathcal{B}_h}(A_h x)\right) = \frac{a_k(\sigma_h; s_h)}{s_h^k}\, \Pi_k Q_{h,k}.$$

Thus, when $a_k(\sigma_h; s_h) \neq 0$, the activation only rescales the degree-k direction $\Pi_k Q_{h,k}$ selected by that branch. Since the coefficients θ_h are independent, varying θ therefore gives exactly

$$\{\Pi_k F_\theta : \theta \in \mathbb{R}^M\} = \operatorname{span}\{\Pi_k Q_{h,k} : 1 \le h \le M\} = \Pi_k \mathcal{S}_k.$$

It remains to determine the dimension of this span. The matrix $\mathsf{G}^{(k)}$ is precisely the Gram matrix of the vectors $\Pi_k Q_{h,k}$. The same calculation as in Theorem 5.1.6 gives

$$\mathsf{G}^{(k)}_{h\ell} = k!\ \operatorname{tr}\left(R_h (A_h A_\ell^*)^{\circ k} R_\ell^*\right), \quad \text{and} \quad \operatorname{rank} \mathsf{G}^{(k)} = \dim \Pi_k \mathcal{S}_k.$$

Finally, by Lemma 3.4.2, Π_k is an isomorphism on homogeneous degree-k polynomials, so this accessible space equals the full degree-k equivariant polynomial space exactly when

$$\operatorname{rank} \mathsf{G}^{(k)} = \dim \operatorname{Hom}_G(\operatorname{Sym}^k L, K).$$

If $a_k(\sigma_h; s_h) = 0$, the corresponding branch contributes nothing in degree k and may simply be omitted. □

This is the degree-k space accessible for the chosen lifts A_h and readouts R_h. Varying those maps may enlarge it. Moreover, although each degree can be analysed separately, the same coefficients θ_h act across all degrees, so the degree components cannot in general be chosen independently.

This also yields an approximation obstruction. For $T \in L^2(\gamma_L; K)^G$ and any degree bound $D \ge 0$,

$$\inf_\theta \|T - F_\theta\|_{L^2}^2 \ge \sum_{k=0}^{D} \operatorname{dist}_{L^2}\left(\Pi_k T, \Pi_k \mathcal{S}_k\right)^2,$$

since the degree-k component of F_θ lies in $\Pi_k \mathcal{S}_k$. Thus any component of $\Pi_k T$ outside this space gives an unavoidable error for the fixed architecture under the Gaussian reference measure.

To control several initial degrees independently, we next construct activations with prescribed Hermite coefficients.

Proposition 6.4.2. *Fix $D \geq 0$. There exist real $t_0, \dots, t_D$ such that, for every $(b_0, \dots, b_D) \in \mathbb{R}^{D+1}$, there are $\theta_0, \dots, \theta_D \in \mathbb{R}$ for which the continuous piecewise linear function*

$$\sigma(z) = \sum_{r=0}^{D} \theta_r \operatorname{ReLU}(z - t_r)$$

satisfies $a_k(\sigma; 1) = b_k$ for $0 \leq k \leq D$. Thus the first $D + 1$ Hermite coefficients can be chosen independently.

Proof. For $0 \leq k \leq D$, set $f_k(t) := a_k(\operatorname{ReLU}(\cdot - t))$. By linearity of the Hermite coefficients, if $\sigma(z) = \sum_{r=0}^{D} \theta_r \operatorname{ReLU}(z - t_r)$, then

$$a_k(\sigma; 1) = \sum_{r=0}^{D} \theta_r f_k(t_r).$$

It is therefore enough to show that $(f_0(t), \dots, f_D(t))$, as t varies, spans $\mathbb{R}^{D+1}$. If not, there exist coefficients $c_0, \dots, c_D$, not all zero, such that $\sum_{k=0}^{D} c_k f_k(t) = 0$ for every t. Differentiating $f_k(t)$ twice gives $f_k''(t) = \frac{1}{k!}\operatorname{He}_k(t)\varphi(t)$. Hence

$$0 = \sum_{k=0}^{D} c_k f_k''(t) = \varphi(t) \sum_{k=0}^{D} \frac{c_k}{k!} \operatorname{He}_k(t).$$

Since $\varphi(t) > 0$ and the Hermite polynomials are linearly independent, $c_k = 0$ for every k.

Therefore the vectors $(f_0(t), \dots, f_D(t))$ span $\mathbb{R}^{D+1}$. Choose $t_0, \dots, t_D$ giving a basis. Then, for any prescribed $(b_0, \dots, b_D)$, suitable coefficients θ_r give $a_k(\sigma; 1) = b_k$ for all $0 \leq k \leq D$. □

For a lift with coordinate variance s^2, the preceding proposition may be applied after rescaling the hidden coordinates by s. Choosing $b_k = \delta_{kj}$ produces a piecewise linear activation whose chaos components vanish in every degree $0, \dots, D$ except degree j. Components above degree D may still be present.

Two polynomial constructions give exact control of chaos degree or homogeneous degree. Using the polynomial activation He_j coordinatewise gives the pure jth-chaos component

$$R(\operatorname{He}_j)_{\mathscr{B}}(Ax/s) = s^{-j} \Pi_j \left[R\left((Ax)^{\odot j} \right) \right].$$

Alternatively, $R\left((Ax)^{\odot j}\right)$ is itself an ordinary homogeneous polynomial of degree j.

Thus the piecewise linear construction controls only finitely many chaos coefficients, whereas the Hermite activation isolates a single chaos degree exactly. In particular, Proposition 6.4.2 does not control the higher-degree tail.

6.5. Regular lifts span every degree

Regular representations of finite groups give a simple construction showing that coordinatewise powers can realise every equivariant polynomial map of a fixed degree. The obstruction of Example 5.4.1 is therefore a property of the chosen coordinates, not an intrinsic one.

Proposition 6.5.1. *Let G be finite and let L, K be real orthogonal G-representations. For $u \in L$ define the linear map*

$$A_u : L \longrightarrow \mathbb{R}[G], \qquad (A_u x)_g = \langle x, gu \rangle ,$$

where $\mathbb{R}[G]$ carries the left regular action, and for $v \in K$ define $A_v : K \to \mathbb{R}[G]$ analogously and put $R_v := |G|^{-1} A_v^*$, that is, $R_v z = |G|^{-1} \sum_{g \in G} z_g \, gv$. Then A_u and R_v are equivariant, their coordinate polynomial is

$$Q_{u,v,k}(x) = R_v \left((A_u x)^{\odot k} \right) = \frac{1}{|G|} \sum_{g \in G} \langle x, gu \rangle^k \, gv,$$

and for every $k \geq 0$ these polynomials span $\mathrm{Poly}_k(L, K)^G$ as u and v vary.

Proof. Equivariance of A_u is the identity $(A_u(hx))_g = \langle x, h^{-1} gu \rangle = (A_u x)_{h^{-1} g}$, and R_v is a multiple of the adjoint of an equivariant map, hence equivariant. Since $A_v^* e_g = gv$, the displayed formula for $Q_{u,v,k}$ follows.

Pure powers of linear forms span the homogeneous scalar polynomials of degree k, so the maps $P_{u,v}(x) = \langle x, u \rangle^k \, v$ span $\mathrm{Poly}_k(L, K)$. Averaging, that is,

$$\mathscr{R}(P)(x) = |G|^{-1} \sum_g g P(g^{-1} x),$$

projects $\mathrm{Poly}_k(L, K)$ onto $\mathrm{Poly}_k(L, K)^G$, and orthogonality of the action gives $\mathscr{R}(P_{u,v}) = Q_{u,v,k}$. A projection maps a spanning set of the ambient space to a spanning set of its image. □

The proposition gives a universal, but generally inefficient, way to match any prescribed equivariant polynomial components through a finite degree bound D. For the normalised construction, assume $L \neq 0$; if $L = 0$, only constant maps remain. For each required degree k, choose finitely many $Q_{u,v,k}$ spanning the desired component, with u normalised so that the coordinates of $A_u X$ have variance one.

By Proposition 6.4.2, one may choose a piecewise linear activation whose Hermite coefficients vanish in all degrees $0, \dots, D$ except degree k. Then (8) shows that, through degree D, the corresponding layer contributes only the chaos projection of the chosen degree-k coordinate polynomial. Summing over the required degrees gives the prescribed chaos components.

Thus copies of the regular representation are sufficient to recover every equivariant polynomial direction. The cost is that each regular block has $|G|$ hidden coordinates, and spanning several directions may require several such blocks.

Chapter 7

Subset representations of the symmetric group

We now make the ambient polynomial spaces from Chapter 4 explicit for an important family of permutation representations. Subset representations model vertex, edge and higher-order features in equivariant graph networks. This chapter describes the full polynomial spaces against which the coordinate-selected spaces of Chapter 5 and Section 6.4 can be compared.

Definition 7.0.1. Let $[n] = \{1, \dots, n\}$, with $[0] = \emptyset$, and let S_n be the symmetric group on $[n]$. For $r \geq 0$, define

$$X_{n,r} := \binom{[n]}{r} = \{A \subseteq [n] : |A| = r\}.$$

The *r-subset representation* is the real vector space

$$V_{n,r} := \mathbb{R}[X_{n,r}] = \bigoplus_{A \in X_{n,r}} \mathbb{R} e_A,$$

equipped with the inner product for which the basis $\{e_A : A \in X_{n,r}\}$ is orthonormal. The action of S_n is defined by $g \cdot e_A = e_{gA}$, where $gA = \{g(a) : a \in A\}$.

The space $V_{n,0} \cong \mathbb{R}$ is the trivial representation. Intuitively, $V_{n,1}$ consists of vertex features, $V_{n,2}$ of features on unordered pairs of distinct vertices, and $V_{n,3}$ of features on unordered triples. Pair features may be viewed as real edge weights. Triple features are assigned to all triples, whether or not those triples form triangles in a particular graph. Throughout, these spaces contain arbitrary real-valued features on every subset of the indicated size.

Fix an input subset size $r \geq 1$, an output subset size $s \geq 0$, and a polynomial degree $d \geq 0$. Our objective is to construct an explicit basis of

$$\mathrm{Poly}_d(V_{n,r}, V_{n,s})^{S_n} \cong \mathrm{Hom}_{S_n}(\mathrm{Sym}^d V_{n,r}, V_{n,s}).$$

An element of this space has the form

$$P(x) = \sum_{Q \in X_{n,s}} P_Q(x) e_Q,$$

where each P_Q is a homogeneous polynomial of degree d in the input coordinates x_A. Equivariance means $P(g \cdot x) = g \cdot P(x)$, or equivalently, $P_{gQ}(g \cdot x) = P_Q(x)$ for $g \in S_n$, $Q \in X_{n,s}$.

A degree-d input monomial $x_{A_1} \cdots x_{A_d}$ records d input subsets, with repetitions allowed. An output coordinate also specifies an s-subset Q. The basis constructed below is indexed by these configurations up to simultaneous relabelling of the vertices. Each basis map sums the monomials belonging to one such configuration type.

7.1. Symmetric powers are multihypergraph modules

To describe symmetric powers combinatorially, for a finite G-set X and $d \geq 0$, let $\mathrm{MSet}_d(X)$ denote the size-d multisets on X with the induced action, which we identify with functions $m : X \to \mathbb{N}_0$ with $\sum_{x \in X} m(x) = d$.

Proposition 7.1.1. *For a finite group G and a finite G-set X,*

$$\mathrm{Sym}^d(\mathbb{R}[X]) \cong \mathbb{R}[\mathrm{MSet}_d(X)]$$

as G-representations. If m_ω represents an orbit $\omega \in \mathrm{MSet}_d(X)/G$ and $H_\omega = \mathrm{Stab}_G(m_\omega)$, then

$$\mathrm{Sym}^d(\mathbb{R}[X]) \cong \bigoplus_{\omega \in \mathrm{MSet}_d(X)/G} \mathrm{Ind}_{H_\omega}^G \mathbf{1}.$$

Proof. The monomials $e^m = \prod_{x \in X} e_x^{m(x)}$ for $m \in \mathrm{MSet}_d(X)$ form a basis of $\mathrm{Sym}^d(\mathbb{R}[X])$, and $ge^m = e^{gm}$. Each orbit therefore spans a permutation module isomorphic to $\mathbb{R}[G/H_\omega] = \mathrm{Ind}_{H_\omega}^G \mathbf{1}$. □

Definition 7.1.2. An *r-uniform multihypergraph on $[n]$ with d edges* is a multiset

$$\mathscr{H} = \{E_1, \dots, E_d\}, \qquad E_i \in X_{n,r},$$

where repeated edges are allowed.

- A vertex is *active* if it lies in at least one edge of $\mathscr{H}$. The subhypergraph obtained by deleting all inactive vertices is called the *active core*.
- Two multihypergraphs are *isomorphic* if there is a bijection between their vertex sets that preserves edge multiplicities. We write $\mathrm{Aut}(\mathscr{H})$ for the automorphism group of $\mathscr{H}$.
- Let $\mathfrak{H}_{r,d}$ denote the set of isomorphism classes of r-uniform multihypergraphs with d edges and no inactive vertices. For $\mathscr{H} \in \mathfrak{H}_{r,d}$, write $v(\mathscr{H})$ for its number of vertices.

Take $X = X_{n,r}$. A monomial $e_{A_1} \cdots e_{A_d}$ corresponds to the edge multiset $\{A_1, \dots, A_d\}$, and two monomials lie in the same S_n-orbit exactly when their active cores are isomorphic. Every core satisfies $v(\mathscr{H}) \leq rd$, with equality precisely when its edges are pairwise disjoint.

For $v(\mathscr{H}) \leq n$, let $P_{\mathscr{H}}(n)$ denote the span of all monomials whose active core is isomorphic to $\mathscr{H}$.

The active core is exactly the orbit data, so it gives the desired decomposition of the symmetric power.

Theorem 7.1.3. *For $n \geq 0$, $r \geq 1$ and $d \geq 0$,*

$$\mathrm{Sym}^d V_{n,r} = \bigoplus_{\substack{[\mathscr{H}] \in \mathfrak{H}_{r,d} \\ v(\mathscr{H}) \leq n}} P_{\mathscr{H}}(n), \qquad \text{where} \qquad P_{\mathscr{H}}(n) \cong \mathrm{Ind}_{\mathrm{Aut}(\mathscr{H}) \times S_{n - v(\mathscr{H})}}^{S_n} \mathbf{1}.$$

The subgroup acts by automorphisms on a labelled copy of the core and by arbitrary permutations on its complement. For $n \geq rd$, every core type occurs.

For each partition $\lambda \vdash n$, the multiplicity of the irreducible Specht module S^λ is

$$[P_{\mathscr{H}}(n) : S^\lambda] = \dim(S^\lambda)^{\mathrm{Aut}(\mathscr{H}) \times S_{n-v(\mathscr{H})}}.$$

Proof. Write $v = v(\mathscr{H})$. By Proposition 7.1.1, the monomials e^m with $m \in \mathrm{MSet}_d(X_{n,r})$ form a permutation basis of $\mathrm{Sym}^d V_{n,r}$, and e^m is identified with the multihypergraph having $m(A)$ copies of each edge A. A permutation maps the active core of m isomorphically onto that of gm, and conversely any isomorphism of active cores extends arbitrarily to the inactive vertices; hence orbits correspond to isomorphism classes of cores. The stabiliser of a monomial with core $\mathscr{H}$ consists of the automorphisms of the core together with arbitrary permutations of the $n - v(\mathscr{H})$ inactive vertices, which identifies $P_{\mathscr{H}}(n)$. Since $v(\mathscr{H}) \le rd$, every core type occurs once $n \ge rd$. Finally, the real Specht modules are absolutely irreducible, so Frobenius reciprocity gives $[P_{\mathscr{H}}(n) : S^\lambda] = \dim \mathrm{Hom}_{\mathrm{Aut}(\mathscr{H}) \times S_{n-v}}(\mathbf{1}, \mathrm{Res}^{S_n}_{\mathrm{Aut}(\mathscr{H}) \times S_{n-v}} S^\lambda) = \dim(S^\lambda)^{\mathrm{Aut}(\mathscr{H}) \times S_{n-v}}$. □

Only the orbit indexing is needed for the polynomial basis below. Representation-stability consequences of Theorem 7.1.3 are recorded in Appendix B.

Example 7.1.4. For $r = 2$, $d = 2$ and $n \ge 4$, there are three core types: a repeated edge, two adjacent edges and two disjoint edges. Writing $e_{ab} = e_{\{a,b\}}$, the representatives are e_{12}^2, $e_{12}e_{23}$ and $e_{12}e_{34}$, respectively. Hence

$$\mathrm{Sym}^2 V_{n,2} = P_{\mathrm{rep}}(n) \oplus P_{\mathrm{adj}}(n) \oplus P_{\mathrm{disj}}(n).$$

Figure 7.1 shows the three cores of Example 7.1.4 and the corresponding orbit modules.

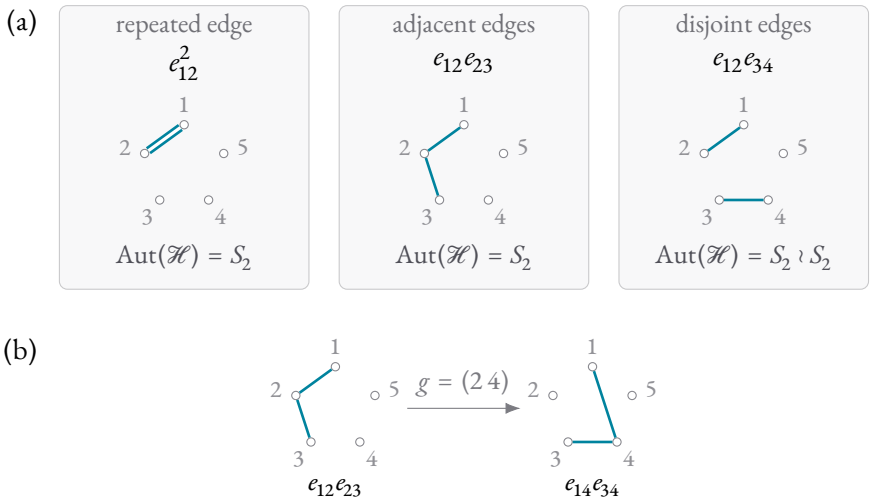


FIGURE 7.1. (a) The three active cores of monomials in $\mathrm{Sym}^2 V_{n,2}$, drawn for $n = 5$. The S_n-orbit of a monomial is determined by its core $\mathscr{H}$ and spans $P_{\mathscr{H}}(n) \cong \mathrm{Ind}^{S_n}_{\mathrm{Aut}(\mathscr{H}) \times S_{n-v(\mathscr{H})}} \mathbf{1}$ (Theorem 7.1.3). (b) Relabelling vertices moves a monomial within its orbit.

7.2. Polynomial equivariants: the marked-orbit basis

The marked-orbit basis describes all homogeneous polynomial maps from r-subset to s-subset features. We begin with the classical linear case.

Proposition 7.2.1. *For finite G-sets X, Y we have* $\mathrm{Hom}_G(\mathbb{R}[X], \mathbb{R}[Y]) \cong \mathbb{R}[X \times Y]^G$, *with basis the maps*

$$T_{\mathcal{O}}(e_x) = \sum_{y:(x,y)\in\mathcal{O}} e_y$$

indexed by the G-orbits $\mathcal{O} \subseteq X \times Y$.

Proof. A linear map T with matrix $K_T(x, y)$ is equivariant if and only if $K_T(gx, gy) = K_T(x, y)$, that is, if and only if K_T is constant on diagonal orbits. See also [GTW25, Lemma 2E.3] and [Mar+19]. □

By Proposition 7.1.1 we may apply this with $X = \mathrm{MSet}_d(X_{n,r})$, which leads to the following description of the orbits.

Definition 7.2.2. A *marked multihypergraph* of degree d and sizes (r, s) is a pair $(\mathcal{A}, Q)$ with $\mathcal{A} \in \mathrm{MSet}_d(X_{n,r})$ and $Q \in X_{n,s}$, that is, an r-uniform d-edge multihypergraph on $[n]$ together with a distinguished s-subset of vertices. Its *marked type* is its orbit under simultaneous relabelling. Marked vertices need not be active.

Theorem 7.2.3. $\mathrm{Poly}_d(V_{n,r}, V_{n,s})^{S_n} \cong \mathrm{Hom}_{S_n}(\mathrm{Sym}^d V_{n,r}, V_{n,s})$, *with a basis indexed by the diagonal S_n-orbits $\mathcal{O} \subseteq \mathrm{MSet}_d(X_{n,r}) \times X_{n,s}$. For such an orbit, define*

$$\Psi_{\mathcal{O}}(x) := \sum_{Q\in X_{n,s}} \left(\sum_{\mathcal{A}:(\mathcal{A},Q)\in\mathcal{O}} x^{\mathcal{A}} \right) e_Q, \qquad x^{\mathcal{A}} := \prod_{A\in X_{n,r}} x_A^{\mathcal{A}(A)}.$$

Then the maps $\Psi_{\mathcal{O}}$ form a basis of $\mathrm{Poly}_d(V_{n,r}, V_{n,s})^{S_n}$.

Proof. By Proposition 7.1.1, a degree-d monomial is indexed by a multiset $\mathcal{A} \in \mathrm{MSet}_d(X_{n,r})$. Hence a polynomial map $V_{n,r} \to V_{n,s}$ is determined by coefficients $c_{\mathcal{A},Q}$ indexed by $(\mathcal{A}, Q) \in \mathrm{MSet}_d(X_{n,r}) \times X_{n,s}$. By Proposition 7.2.1, such a map is S_n-equivariant exactly when these coefficients are constant on diagonal S_n-orbits. The corresponding orbit indicators are precisely the maps $\Psi_{\mathcal{O}}$. □

Under the polarisation identification of Proposition 2.4.3, the usual multinomial factors only rescale the multiset basis vectors. These factors are constant on S_n-orbits, so they do not change the orbit indexing or the basis statement.

Orbit-indexed graph-polynomial bases also appear in [Pun+23]. We next determine the dimension and stable range of this basis. Fixing the active core reduces the count to the possible positions of the output mark.

Proposition 7.2.4. *Fix an active core $\mathcal{H}$ with vertex set $U = V(\mathcal{H})$, and write $v = v(\mathcal{H})$. A marked type with core $\mathcal{H}$ is determined by $q := |Q \cap U|$ together with the* $\mathrm{Aut}(\mathcal{H})$*-orbit of the*

active part $Q \cap U \in \binom{U}{q}$. *Consequently,*

$$\dim \mathrm{Hom}_{S_n}(\mathrm{Sym}^d V_{n,r}, V_{n,s}) = \sum_{\substack{[\mathscr{H}]\in\mathfrak{H}_{r,d} \\ v(\mathscr{H})\le n}} \sum_{q=\max(0,\, s-(n-v(\mathscr{H})))}^{\min(s,\, v(\mathscr{H}))} \left| \binom{U}{q} \Big/ \mathrm{Aut}(\mathscr{H}) \right|. \tag{9}$$

Proof. By Theorem 7.1.3 and Frobenius reciprocity, the contribution of a fixed core $\mathscr{H}$ is

$$\dim (V_{n,s})^{\mathrm{Aut}(\mathscr{H})\times S_{n-v}}.$$

Since $V_{n,s} = \mathbb{R}[X_{n,s}]$ is a permutation representation, this dimension is the number of $(\mathrm{Aut}(\mathscr{H}) \times S_{n-v})$-orbits on s-subsets $Q \subseteq [n]$.

Write $q = |Q \cap U|$. $\mathrm{Aut}(\mathscr{H})$ acts on the active part $Q \cap U$, while S_{n-v} is transitive on all $(s-q)$-subsets of the inactive vertices. Thus, for fixed q, the orbit of Q is determined exactly by the $\mathrm{Aut}(\mathscr{H})$-orbit of $Q \cap U \in \binom{U}{q}$.

Such a marking exists precisely when $0 \le q \le v$ and $0 \le s - q \le n - v$, giving the stated limits. Summing over the core types proves (9). □

The counting formula also shows exactly when increasing n can no longer create new marked types. This is a statement about sharing coefficients across graph orders. Compatibility with embeddings or other feature-transfer maps is an additional requirement; see [LD24].

Theorem 7.2.5. *Let* $r, d \ge 1$ *and* $s \ge 0$, *and write* $D(n) := \dim \mathrm{Hom}_{S_n}(\mathrm{Sym}^d V_{n,r}, V_{n,s})$. *Then:*

(a) $D(n)$ *is non-decreasing in* n.

(b) *For* $n \ge rd + s$,

$$D(n) = \sum_{[\mathscr{H}]\in\mathfrak{H}_{r,d}} \sum_{q=0}^{\min(s, v(\mathscr{H}))} \left| \binom{V(\mathscr{H})}{q} \Big/ \mathrm{Aut}(\mathscr{H}) \right|,$$

so $D(n)$ *is independent of* n *in this range.*

(c) *The threshold is sharp:* $D(rd + s - 1) < D(rd + s)$.

In the stable range, the basis $\{Y_{\mathscr{O}}\}$ *is indexed by a set independent of* n, *so one coefficient vector defines one equivariant polynomial layer simultaneously for every* $n \ge rd + s$.

Proof. In (9) the set of admissible types grows with n and each lower summation limit is non-increasing in n, so D is non-decreasing. For $n \ge rd + s$ we have $n - v \ge s$ for every core, so every lower limit is 0 and no term depends on n, which gives *(b)*.

For sharpness, let $\mathscr{H}_0$ be the disjoint type, with $v(\mathscr{H}_0) = rd$. At $n = rd + s - 1$ the value $q = 0$ would require $s - q \le n - v = s - 1$, which fails, so $q = 0$ is inadmissible for $\mathscr{H}_0$. Also, when $s = 0$, the core $\mathscr{H}_0$ does not embed at all. At $n = rd + s$ it becomes admissible and contributes exactly one further orbit, while no other term decreases. Hence the strict inequality. □

7.3. Worked example: edge-to-vertex maps

Take $(r, d, s) = (2, 2, 1)$ and write $x_{uv} = x_{\{u,v\}}$. By Theorem 7.2.3 we must list the marked types: a repeated edge with the mark on it or off it; an adjacent pair with the mark at the centre, at an endpoint or outside; a disjoint pair with the mark on an active vertex or outside. These give seven types (Figure 7.2), hence $\dim \mathrm{Hom}_{S_n}(\mathrm{Sym}^2 V_{n,2}, V_{n,1}) = 7$ for $n \geq 5$. At $n = 3$ only four types fit, while at $n = 4$ only the disjoint pair with an outside mark is absent. Thus the dimensions are $4, 6, 7, 7, \ldots$ for $n = 3, 4, 5, 6, \ldots$.

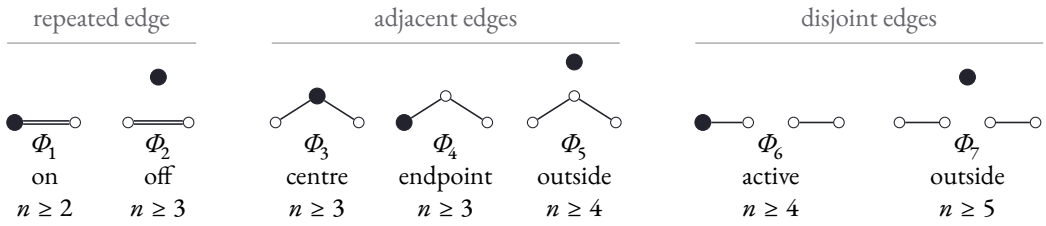


Figure 7.2. The seven marked types for quadratic maps $V_{n,2} \to V_{n,1}$, grouped by active core. Filled vertices carry the output mark v, and double lines denote a repeated input edge. Each type is labelled by the least n at which it occurs.

Writing the vth output coordinate gives the following basis, using the same labels as Figure 7.2.

$$
\begin{aligned}
(\Phi_1 x)_v &= \sum_{u \neq v} x_{uv}^2, && \text{repeated edge, mark on,}\\
(\Phi_2 x)_v &= \sum_{\{u,w\} \not\ni v} x_{uw}^2, && \text{repeated edge, mark off,}\\
(\Phi_3 x)_v &= \sum_{\{a,c\} \subseteq [n] \setminus \{v\}} x_{av} x_{vc}, && \text{adjacent edges, centre,}\\
(\Phi_4 x)_v &= \sum_{u \neq v} \sum_{w \notin \{u,v\}} x_{vu} x_{uw}, && \text{adjacent edges, endpoint,}\\
(\Phi_5 x)_v &= \sum_{b \neq v} \sum_{\{a,c\} \subseteq [n] \setminus \{v,b\}} x_{ab} x_{bc}, && \text{adjacent edges, outside,}\\
(\Phi_6 x)_v &= \sum_{u \neq v} \sum_{\{a,b\} \subseteq [n] \setminus \{v,u\}} x_{vu} x_{ab}, && \text{disjoint edges, active,}\\
(\Phi_7 x)_v &= \sum_{\substack{\{E_1,E_2\} \\ E_1 \cap E_2 = \emptyset \\ v \notin E_1 \cup E_2}} x_{E_1} x_{E_2}, && \text{disjoint edges, outside.}
\end{aligned}
$$

In the last sum, E_1 and E_2 are two-element subsets of $[n]$ and their pair is unordered. Each map is an orbit sum of one marked type. Their monomial supports are disjoint, so the maps are independent and the dimension count shows that they form a basis for $n \geq 5$.

The marked types give each basis map a direct interpretation. Φ_1 is a local square statistic at v, while Φ_3 and Φ_4 are quadratic contractions along length-two paths through or ending at v. The remaining four maps involve configurations that are not local path contractions at v.

Together with the equivariant linear maps $V_{n,2} \to V_{n,1}$ and the constants, the seven maps span all equivariant polynomial maps $V_{n,2} \to V_{n,1}$ of degree at most two. This example describes the full ambient quadratic space. The next chapter instead fixes a coordinatewise construction on triple features and uses the Gram criterion to determine which source-to-target blocks it produces.

Chapter 8

A representation-aware architecture on graph triples

We now use the coordinatewise theory to organise a graph network by irreducible source and target, where features are indexed by unordered triples of vertices. For $n \geq 6$, the triple module has four irreducible components, interpreted as global, vertex-level, pair-level and residual triple-level variation. We determine its ReLU routes and test this organisation as an inductive bias on a synthetic benchmark.

8.1. Route-resolved coordinatewise layers

Let $H = \mathbb{R}^{\mathscr{B}}$ be a transitive permutation representation with a multiplicity-free decomposition $H = \bigoplus_{i=0}^{m} L_i$ into irreducibles, where L_0 is the trivial summand. As in Section 5.3, write E_i for the orthogonal projector onto L_i, $d_i = \dim L_i$, and

$$s_i^2 = \frac{d_i}{\dim H}$$

for the common variance of the coordinates of a standard Gaussian vector in L_i (Lemma 5.1.3). For an activation σ, the *route* from L_i to L_j is the normalised block

$$F_{j\leftarrow i} : L_i \longrightarrow L_j, \qquad F_{j\leftarrow i}(x) = p_j\, \sigma_{\mathscr{B}}\big(s_i^{-1} x\big).$$

This is the coordinatewise layer of Theorem 5.1.6 with lift $A = s_i^{-1}\iota_i$ and readout $R = p_j$. Under the Gaussian reference input, the factor s_i^{-1} gives every hidden coordinate unit variance, so all routes use the same reference scale. Theorem 5.1.6 then reads

$$\Pi_k F_{j\leftarrow i} = \frac{a_k(\sigma)}{s_i^k}\, \Pi_k Q_k^{ij}, \qquad Q_k^{ij}(x) = p_j\big(x^{\odot k}\big), \tag{10}$$

and by Corollary 5.3.2, $Q_k^{ij} \neq 0$ if and only if the projector coefficient $\kappa_j^{(i)}(k) := d_j^{-1}\operatorname{tr}(E_j E_i^{\circ k})$ is positive. Since $F_{j\leftarrow i}$ is the block $(\sigma_{\mathscr{B}})_i^j$ precomposed with the invertible input rescaling $x \mapsto s_i^{-1}x$, the routes which are not identically zero are exactly the edges of the interaction graph $\Gamma_{\sigma_{\mathscr{B}}}$. We call the set of index pairs (i, j) for which $F_{j\leftarrow i} \not\equiv 0$, that is, of edges $L_i \to L_j$ of $\Gamma_{\sigma_{\mathscr{B}}}$, the *route set* of σ on H.

For ReLU, the route set is determined by the projector coefficients in its supported degrees.

Lemma 8.1.1. *Let σ be positively homogeneous with $m_+ \neq \pm m_-$, for example $\sigma = \mathrm{ReLU}$. Then (i, j) lies in the route set of σ on H if and only if one of the following holds:*

(a) $j = i$, *or* $j = 0$;

(b) $i \neq 0$ *and* $\kappa_j^{(i)}(2r) > 0$ *for some* $r \geq 1$.

If $\kappa_j^{(i)}(2) > 0$, *then in addition* $\kappa_j^{(i)}(2r) > 0$ *for every* $r \geq 1$.

Proof. By Theorem 6.1.2, $\operatorname{supp} a_\bullet(\sigma) = \{1\} \cup 2\mathbb{N}_0$, so by (10) and Proposition 5.5.2 the route is nonzero exactly when $\kappa_j^{(i)}(k) > 0$ for some $k \in \{0, 1\} \cup \{2, 4, 6, \dots\}$. Since $E_i^{\circ 0} = J = (\dim H)E_0$ and $E_i^{\circ 1} = E_i$, we have $\kappa_j^{(i)}(0) > 0$ if and only if $j = 0$, and $\kappa_j^{(i)}(1) > 0$ if and only if $j = i$. If $i = 0$, then $x \in L_0$ is a constant vector, so $x^{\odot k} \in L_0$ and $\kappa_j^{(0)}(k) = 0$ for $j \neq 0$. The final statement is Lemma 5.3.3. □

The features of the implementation below directly fit into this framework. First, positive channel gains $g > 0$ do not affect the route set of a homogeneous activation, since $\sigma(gz) = g\sigma(z)$. Second, features usually carry c channels on which G acts trivially, so the feature space is $H \otimes \mathbb{R}^c$ with hidden basis $\mathcal{B} \times \{1, \dots, c\}$, which has c coordinate orbits. When the projectors act as $E_i \otimes I_c$ and the activation is applied coordinatewise, the channels do not interact, and each channel is a copy of the layer above. Channel mixing before the activation would instead require the several-orbit formula of Remark 5.1.8.

8.2. The triple module

We now specialise to subset modules $V_{n,r} = \mathbb{R}[X_{n,r}]$ of Chapter 7.

Proposition 8.2.1. *For* $0 \leq r \leq n$, $V_{n,r} \cong \operatorname{Ind}_{S_r \times S_{n-r}}^{S_n} \mathbf{1}$. *For* $r \leq n/2$,

$$V_{n,r} \cong \bigoplus_{i=0}^{r} S^{(n-i,i)}, \qquad \dim S^{(n-i,i)} = \binom{n}{i} - \binom{n}{i-1}.$$

In particular $V_{n,r}$ *is transitive and multiplicity-free.*

Proof. The action on r-subsets is transitive with stabiliser $S_r \times S_{n-r}$. The decomposition of this Young permutation module is Young's rule [Jam78; Sag01]. The dimension formula follows by induction on r. □

Hence Section 5.3 and Section 8.1 apply to $V_{n,r}$. A pair (T, T') of r-subsets is determined up to simultaneous relabelling by $t = |T \setminus T'|$, so the matrices

$$(A_t)_{T,T'} = \begin{cases} 1, & |T \setminus T'| = t, \\ 0, & \text{otherwise}, \end{cases} \qquad 0 \leq t \leq r,$$

form the orbit basis of $\operatorname{End}_{S_n}(V_{n,r})$ given by Proposition 7.2.1. They span the Bose–Mesner algebra of the Johnson scheme $J(n, r)$, whose primitive idempotents are the projectors E_i onto $S^{(n-i,i)}$ [Del73; BCN89]. In particular every entry of E_i depends only on t, and the Hadamard coefficients of Corollary 5.3.2 are the Krein parameters of the scheme up to normalisation.

From now on $r = 3$ and $n \geq 6$.

Write $N = \binom{n}{3}$ and $L_i = S^{(n-i,i)}$, so that

$$V_{n,3} \cong L_0 \oplus L_1 \oplus L_2 \oplus L_3, \qquad (d_0, d_1, d_2, d_3) = \left(1,\ n-1,\ \tfrac{n(n-3)}{2},\ \tfrac{n(n-1)(n-5)}{6}\right). \tag{11}$$

We read L_0 as global variation, L_1 as variation attributable to individual vertices, L_2 as residual pair-level variation and L_3 as genuinely triple-level variation; this is an interpretation of the filtration $V_{n,0} \subseteq V_{n,1} \subseteq V_{n,2} \subseteq V_{n,3}$ obtained by pulling back vertex and pair features to triples.

The adjacency matrix A_1 of the Johnson graph has eigenvalues coming from the Eberlein polynomial

$$\theta_i = P_1(i) = (3-i)(n-3-i) - i, \qquad (\theta_0, \theta_1, \theta_2, \theta_3) = (3n-9,\ 2n-9,\ n-7,\ -3),$$

which are pairwise distinct for $n \geq 6$. Consequently

$$E_i = \prod_{l \neq i} \frac{A_1 - \theta_l I}{\theta_i - \theta_l}, \tag{12}$$

and these polynomials in the sparse matrix A_1 give exact projectors for every $n \geq 6$. The explicit entries and their derivation are recorded in Table C.1 and Appendix C.

Since each E_i is symmetric with entries depending only on t, the projector coefficients are finite sums:

$$\kappa_j^{(i)}(k) = \frac{1}{d_j} \operatorname{tr}\left(E_j E_i^{\circ k}\right) = \frac{N}{d_j} \sum_{t=0}^{3} v_t \, (E_j)_t \, (E_i)_t^k, \tag{13}$$

where $(E_i)_t$ denotes the common value of the entries at $\left|T \setminus T'\right| = t$, and $v_t = P_t(0)$ is the valency of A_t.

We can now determine exactly which of the sixteen routes $F_{j \leftarrow i}$ of the triple module are nonzero for ReLU, and in which degree each first appears.

Proposition 8.2.2. *Let σ be positively homogeneous with $m_+ \neq \pm m_-$, for example ReLU, and consider the routes $F_{j \leftarrow i}$ on $V_{n,3}$ for $n \geq 6$.*

(a) *If $n \geq 7$, the route set is*

$$\mathscr{R} = \{(0,0)\} \cup \{(i,j) : 1 \leq i \leq 3,\ 0 \leq j \leq 3\},$$

consisting of thirteen routes. Each route (i,j) with $i \neq j$ is realised in degree 0 or 2, except $L_1 \to L_3$, for which $Q_2^{(1,3)} = 0 \neq Q_4^{(1,3)}$, so that this route first appears in degree four.

(b) *If $n = 6$, the four routes $L_1 \to L_3$, $L_2 \to L_1$, $L_2 \to L_3$ and $L_3 \to L_1$ vanish identically, and the remaining nine routes of $\mathscr{R}$ are nonzero. The routes $L_2 \to L_1$ and $L_2 \to L_3$ vanish for every activation, whereas $L_1 \to L_3$ and $L_3 \to L_1$ are restored by every biased ReLU σ_c with $c \neq 0$.*

Proof. The projector-coefficient calculation and the complementation argument for $n = 6$ are given in Appendix C. □

The order $n = 6$ therefore exhibits two of the three mechanisms of Section 1.2 in one representation, mechanisms which Example 5.4.1 and Section 6.3 exhibit separately. The routes $L_2 \to L_1$ and $L_2 \to L_3$ are excluded by the coordinates: every coordinatewise function of an input in L_2 is fixed by complementation, so changing the activation cannot restore them. The routes between L_1 and L_3 are lost to the activation: their cubic coordinate polynomials are nonzero, and a bias restores them. The obstruction disappears for $n \geq 7$, where complementation no longer maps triples to triples.

Figure 8.1 summarises Proposition 8.2.2 and the three causes.

$n \geq 7$

$L_i \backslash L_j$	L_0	L_1	L_2	L_3
L_0	0	×	×	×
L_1	0	1	2	4
L_2	0	2	1	2
L_3	0	2	2	1

$n = 6$

$L_i \backslash L_j$	L_0	L_1	L_2	L_3
L_0	0	×	×	×
L_1	0	1	2	$3^\dagger$
L_2	0	×	1	×
L_3	0	$3^\dagger$	2	1

× symmetry × coordinates † activation

Figure 8.1. The sixteen routes $L_i \to L_j$ of $V_{n,3}$ for ReLU (Proposition 8.2.2), with source L_i indexing rows and target L_j indexing columns. A number is the first degree in which the route is nonzero. For n = 6, an entry 3† marks a route that vanishes for ReLU and first appears in degree 3 for a biased ReLU.

The degree calculation identifies $L_1 \to L_3$ as a route whose first ReLU component occurs in degree four; all its nonzero ReLU components have even degree. The architecture below retains the complete route rather than separating those degrees.

8.3. Specht Triple

Let $\mathcal{G} = (V, E)$ be a simple graph with $|V| = n$ and adjacency matrix $A_{\mathcal{G}}$. A pretrained, frozen message-passing network, here a Graph Isomorphism Network (GIN) [Xu+19], assigns to every vertex v a hidden state $h_v \in \mathbb{R}^{64}$ and to the graph a baseline prediction $\hat{y}_0$. All three higher-order branches below receive the same triple features and output a residual correction to $\hat{y}_0$.

Triple lift. For $T = \{u, v, w\} \in \binom{V}{3}$, let $c(T) = (A_{\mathcal{G}})_{uv} + (A_{\mathcal{G}})_{uw} + (A_{\mathcal{G}})_{vw} \in \{0, 1, 2, 3\}$ be the number of edges induced on T, and set

$$X_T = \frac{1}{3}\big(W_h h_u + W_h h_v + W_h h_w\big) + e\big(c(T)\big) \in \mathbb{R}^6, \tag{14}$$

where $W_h : \mathbb{R}^{64} \to \mathbb{R}^6$ and the embedding $e : \{0, 1, 2, 3\} \to \mathbb{R}^6$ are learned. Both terms depend only on the unordered set T, and the node states are themselves equivariant, so $X = (X_T)_T \in V_{n,3} \otimes \mathbb{R}^6$ transforms equivariantly, with S_n acting trivially on the six channels. The width 6 is an architectural choice and is not prescribed by the theory.

Specht Triple. For each source i, the branch forms the normalised source activation and evaluates the routes of Section 8.1 channel by channel:

$$V_i = \mathrm{ReLU}\big(g_i \odot s_i^{-1}(E_i \otimes I_6)X\big), \qquad Q_{j\leftarrow i} = (E_j \otimes I_6)V_i, \qquad (i,j) \in \mathscr{R},$$

where $s_i^{-1} = \sqrt{N/d_i}$ and $g_i \in \mathbb{R}^6_{>0}$ are learned channel gains. We identify $V_{n,3} \otimes \mathbb{R}^6$ with $\mathbb{R}^{N\times 6}$, so that $(E_i \otimes I_6)X = E_i X$, as in Algorithm 1. By the remarks after Lemma 8.1.1, each channel of $Q_{j\leftarrow i}$ is the route $F_{j\leftarrow i}$, rescaled by the corresponding gain, and Proposition 8.2.2 shows that $\mathscr{R}$ consists of exactly the routes which are not identically zero, for every $n \geq 7$. The triple index is then removed by the channelwise invariant statistics

$$\Phi(Q) = \Big(\tfrac{1}{N}\textstyle\sum_T Q_T^{\odot 2},\ \tfrac{1}{N}\sum_T \mathrm{ReLU}(Q_T)\Big) \in \mathbb{R}^{12},$$

and the route summaries are accumulated with learned gates $\gamma_{j\leftarrow i} \in \mathbb{R}^{12}$, as in Algorithm 1.

Algorithm 1. Specht Triple residual for one graph with $n \geq 7$ vertices.

Input: adjacency $A_{\mathcal{G}}$, frozen node states $(h_v)_v$ and baseline $\hat{y}_0 \in \mathbb{R}^2$
Output: prediction $\hat{y} \in \mathbb{R}^2$

$X \leftarrow$ triple lift (14), $X \in \mathbb{R}^{N\times 6}$
for $i = 0, 1, 2, 3$ **do**
  $V_i \leftarrow \mathrm{ReLU}\big(g_i \odot \sqrt{N/d_i}\, E_i X\big)$ `// E_i from (12)`
$z \leftarrow 0 \in \mathbb{R}^{12}$
foreach $(i,j) \in \mathscr{R}$ **do**
  $z \leftarrow z + \gamma_{j\leftarrow i} \odot \Phi(E_j V_i)$
return $\hat{y}_0 + \mathrm{Head}\big(\mathrm{LayerNorm}(z/\sqrt{48})\big)$

The learned parameter shapes are independent of n. Recomputing the projectors from (12) defines the branch with the same parameter vector for every $n \geq 7$. This shares parameters across graph orders, as in Theorem 7.2.5; it does not assert compatibility with embeddings between graphs. The architecture does not separate Hermite degrees. The degree calculation is used to determine which source-to-target blocks can be nonzero for ReLU, and those complete blocks are exposed to learning.

Control Triple. To test whether a gain comes from this organisation rather than from access to triples, the first control is given the full linear equivariant mixing space on triple features before its nonlinearity. Let $T_n = \frac{1}{3(n-3)}A_1$ be the normalised Johnson adjacency operator, and define

$$L_J(X) = \sum_{r=0}^{3} T_n^r X W_r, \qquad W_r \in \mathbb{R}^{6\times 6}. \tag{15}$$

The control applies ReLU to $L_J(X)$ (after a fixed rescaling and a learned channel bias), followed by the same statistics Φ, layer normalisation and a residual head.

Lemma 8.3.1. *For $n \geq 6$, the maps $X \mapsto \sum_{r=0}^{3} T_n^r X W_r$ with $W_r \in \mathbb{R}^{6\times 6}$ are exactly the S_n-equivariant linear endomorphisms of $V_{n,3} \otimes \mathbb{R}^6$.*

Proof. By Proposition 8.2.1 and Schur's lemma, $\mathrm{End}_{S_n}(V_{n,3})$ has dimension four, as the four summands are absolutely irreducible and pairwise non-isomorphic. The operator T_n acts on L_i by $\theta_i/(3(n-3))$, and these four values are distinct, so its minimal polynomial has degree four and I, T_n, T_n^2, T_n^3 are linearly independent in this algebra, hence span it. Since S_n acts trivially on the channels, $\mathrm{End}_{S_n}(V_{n,3} \otimes \mathbb{R}^6) = \mathrm{End}_{S_n}(V_{n,3}) \otimes \mathrm{End}(\mathbb{R}^6)$. □

Projector Control. The second control uses the same projectors as Specht Triple, but mixes linearly before a single activation:

$$L_P(X) = \sum_{i=0}^{3} E_i X U_i, \qquad U_i \in \mathbb{R}^{6\times 6}. \tag{16}$$

It applies ReLU to $L_P(X)$ after the same fixed rescaling and learned channel bias as Control Triple, followed by the same statistics Φ, layer normalisation and residual head. The projectors $E_0, \dots, E_3$ are nonzero orthogonal projections onto distinct summands, so they are linearly independent and form a second basis of the four-dimensional algebra $\mathrm{End}_{S_n}(V_{n,3})$. By the proof of Lemma 8.3.1, the maps (16) are therefore again exactly the S_n-equivariant linear endomorphisms of $V_{n,3} \otimes \mathbb{R}^6$. At a fixed order n the two controls realise the same functions; they differ in their parametrisation, and hence in optimisation and in how the weights are shared across graph orders.

In the terminology of Chapter 5, each control is a single coordinatewise layer whose lift is a general equivariant linear map, read out only through invariant statistics. Specht Triple instead applies its activation to each irreducible source separately and keeps every nonzero block $(\sigma_{\mathcal{B}})_i^j$ separate until the invariant readout. By Lemma 8.3.1, both controls have the full linear equivariant mixing space available before activation, and Projector Control moreover has access to the same irreducible projectors as Specht Triple. The comparison of Specht Triple with Projector Control is therefore the primary one: it tests the arrangement of nonlinear computation relative to the irreducible decomposition, with the linear space and the projectors held fixed.

8.4. Experiment

We use a synthetic structural benchmark, so that the targets have clear graph-theoretic meaning and graph orders remain small enough for exact triple representations. Graphs have $15 \le n \le 24$ vertices and are sampled from Erdős–Rényi, Barabási–Albert, Watts–Strogatz and two-block stochastic block model families. The two regression targets are the diameter and the algebraic connectivity, the second-smallest Laplacian eigenvalue, both of which require information beyond a single local neighbourhood. Vertices carry four-dimensional Gaussian random identifiers as input features. For each of ten optimisation seeds, a five-layer GIN is trained and frozen, and the three residual branches, each with exactly 884 trainable parameters, are trained on top of the same checkpoint. Every batch contains graphs of a single order, so the projectors and Johnson operators always match the actual S_n-action. The score is the mean of the two standardised test mean squared errors (MSE). The full protocol is given in Appendix A.

Model	Params.	Mean MSE	Diameter	Alg. conn.
GIN	–	0.09132	0.11720	0.06543
Control Triple	884	0.07640	0.10492	0.04787
Projector Control	884	0.06937	0.10081	0.03794
Specht Triple	884	**0.06171**	**0.08645**	**0.03696**

Table 8.1. Standardised test mean squared errors, averaged over ten paired seeds. The primary comparison is Specht Triple against Projector Control.

Table 8.1 summarises the results. Against the primary control Projector Control, Specht Triple reduces the mean standardised test MSE from 0.06937 to 0.06171, a ratio-of-means reduction of 11.05%, with paired-bootstrap 95% interval [2.24%, 18.27%] and 7/10 seed wins. The exact paired sign-flip test gives a one-sided p-value of 0.0293 and a two-sided value of 0.0586, so the inference is sensitive to sidedness. The gain is uneven across targets: diameter improves by 14.24% (8/10 seeds), whereas the 2.57% improvement in algebraic connectivity (6/10 seeds) remains statistically unresolved. Against the secondary control Control Triple, Specht Triple improves in every one of the ten seeds, with a reduction of 19.23%, paired-bootstrap interval [15.97%, 21.90%] and one-sided sign-flip p-value 2^{-10}, the smallest attainable with ten seeds.

Control Triple already improves the GIN backbone, by 16.34%, which shows that higher-order equivariant processing is useful on this benchmark. Projector Control realises the same linear maps in the projector basis and has a lower mean error than Control Triple (0.06937 against 0.07640). This is consistent with representation-adapted coordinates giving a better-conditioned parametrisation, although we did not test this difference for significance. The further gain of Specht Triple over Projector Control, with the linear space and the projectors held fixed, is consistent with the benefit of organising the nonlinear computation by irreducible source and target. It is, however, not attributable to a single mechanism: the branches still differ in their nonlinear factorisation and in their normalisation, since Specht Triple normalises each source before ReLU and has no pre-activation bias. The gain should therefore be attributed to the representation-aware design as a whole, and the evidence for it is clearest for the diameter.

The branches have equal parameter counts but different computational costs. Specht Triple evaluates thirteen routes. Relative to Control Triple, Specht Triple takes about 4.90 times the incremental training time, 4.02 times the inference time, 4.07 times the peak training memory and 6.26 times the peak inference memory (Table A.3). The frozen backbone limits the comparison to residual branches rather than end-to-end higher-order architectures. Predictions are invariant under relabelling when the random identifiers move with their vertices, but may change when those identifiers are redrawn.

Chapter 9

Conclusion and future directions

This thesis refined the piecewise linear Schur lemma of [GTW25] by asking in which *degree* a nonlinear equivariant interaction occurs and whether a *specified layer* produces it. The Gaussian chaos decomposition supplies the degree, and for coordinatewise layers it distinguishes obstructions from symmetry, hidden coordinates and activation. The resulting picture is explicit for subset features of the symmetric group, and in Chapter 8 it guided a graph architecture that outperformed two parameter-matched equivariant controls on the reported synthetic benchmark, including one with access to the same irreducible projectors.

9.1. Limitations

The chaos transform determines a continuous equivariant map uniquely (Corollary 4.1.2), but it remains open to characterise which sequences of components come from piecewise linear maps. Matching finitely many components with biased ReLUs leaves the higher-degree tail uncontrolled, so it gives no approximation bound by itself. For compact connected groups, polynomial existence does not detect piecewise linear existence (Remark 4.2.4).

The Gram criterion describes single layers, or sums of layers, with fixed lifts and readouts. The one-layer obstructions need not persist under composition, and a description of networks of prescribed width and depth must account for parameters shared across degrees and for interactions created by composition.

The general spanning construction is costly, since each regular hidden block has $|G|$ coordinates, and the multihypergraph orbit modules of Chapter 7 are still reducible in general. The experiment of Chapter 8 uses a frozen backbone, one subset module and one synthetic benchmark. Its primary comparison rests on ten seeds, with a bootstrap interval of [2.24%, 18.27%] and an unresolved gain for algebraic connectivity, and its route-resolved computation has the additional time and memory costs reported in Table A.3.

9.2. Future directions

Constructing architectures

The constructions of Sections 6.4 and 6.5 suggest a design recipe: choose source and target modules and a degree budget, select lifts and readouts whose coordinate polynomials span the intended spaces using the Gram test of Proposition 6.4.1, and choose activations whose Hermite coefficients are nonzero at the required degrees. Specht Triple uses only the coarsest output of this analysis, namely which complete blocks are nonzero. For other representations or choices of activation functions, symmetry and coordinate tests can remove zero routes before evaluation.

The present implementation already omits such routes. Separating degrees within a nonzero route is a further architectural change: Hermite activations isolate an exact chaos degree, whereas biased ReLU combinations control only finitely many coefficients (Proposition 6.4.2). Whether these changes reduce the overhead in Section 8.4 remains to be tested.

Graph networks

The marked-orbit bases of Chapter 7 give every polynomial map between vertex, edge and higher-order features, with coefficients shared across graph orders in the stable range. Combining these polynomial layers with the route-resolved layers of Chapter 8, training end-to-end, and treating other subset sizes are immediate next steps. On Boolean adjacency inputs, identities such as $x_{uv}^2 = x_{uv}$ introduce relations between basis maps which an efficient architecture should exploit.

Triangulated manifolds

A further direction is to classify manifolds from their triangulations. Vertex relabelling is then insufficient, since retriangulations such as Pachner moves change both the vertex set and the incidence structure. Extending the theory requires specifying how features are transferred between triangulations and how layers respect these transfers, so that the final prediction is invariant under the chosen moves.

Appendix A

Protocol for the graph experiment

This appendix records the protocol of the experiment in Section 8.4, following [Sha26b].

A1. Data

A single corpus was generated with seed 440081. Graph orders are uniform on $\{15, \dots, 24\}$, and the corpus is balanced over the four families of Table A.1; disconnected samples are rejected. The split is 8,000/2,000/4,000 graphs for training, validation and testing, with each family contributing 2,000/500/1,000. A four-round Weisfeiler–Leman (1-WL) structural hash excludes overlap between the splits and with 11,981 graphs inspected during development.

Family	Parameters	Range
Erdős–Rényi [ER60]	edge probability p	$p \sim U(0.08, 0.34)$
Barabási–Albert [BA99]	attachments m	$m \in \{1, 2, 3, 4\}$
Watts–Strogatz [WS98]	degree k, rewiring β	$k \in \{2, 4, 6\}$, $\beta \sim U(0.05, 0.45)$
Two-block SBM [HLL83]	block cut, p_{in}, p_{out}	cut in middle third; $U(0.18, 0.55)$; $U(0.02, 0.18)$

TABLE A.1. Parameters of the graph generators, sampled independently for each graph.

Node identifiers are initially drawn independently from $N(0, I_4)$, then centred and scaled coordinatewise within each graph. Targets are the unweighted diameter and the second-smallest eigenvalue of the combinatorial Laplacian $D_{\mathcal{G}} - A_{\mathcal{G}}$, with $D_{\mathcal{G}}$ the degree matrix, each standardised by its training mean and standard deviation, $(4.51775, 1.09949)$ and $(2.33446, 0.79849)$ respectively.

A2. Architectures and parameters

The GIN has depth 5 and width 64, with an input projection $\mathbb{R}^4 \to \mathbb{R}^{64}$, an ε-weighted self term plus adjacency aggregation, a two-layer SiLU multilayer perceptron (MLP) of hidden width 128, a residual connection and layer normalisation in each layer, mean pooling and a prediction head with 64 hidden units. All three residual branches use the lift (14) with width 6. Their trainable parameters are exactly matched, and the two controls agree component by component:

Component	SPECHT TRIPLE	either control
node projection W_b	384	384
edge-count embedding e	24	24
source gains and route gates / mixing and bias	24 + 156	144 + 6
layer normalisation	24	24
residual head	272	302
Total	884	884

The head widths differ only to match the totals. In Specht Triple there are four gain vectors $g_i \in \mathbb{R}^6$ and thirteen gates $\gamma_{j\leftarrow i} \in \mathbb{R}^{12}$. The constant $\sqrt{48}$ in Algorithm 1 was inherited from a 48-route development supernet and retained so as not to change the evaluated checkpoints. In both controls, the mixed tensor (15) or (16) is divided by 2 before adding a learned channel bias, and the head is a $12 \to 20 \to 2$ SiLU network; Specht Triple uses a $12 \to 18 \to 2$ head.

A3. Optimisation and statistics

The GIN is trained for at most 140 epochs at initial learning rate 2×10^{-3}, and each residual branch for at most 140 epochs at 10^{-3}, with batch size 256 (capped at 600,000 triple coordinates), weight decay 10^{-5}, gradient clipping at 5, validation every four epochs, learning-rate halving after three checks without improvement down to 2×10^{-5}, early-stopping patience of twelve checks and mixed precision. Checkpoints are selected by validation score only. Both controls use the frozen GIN checkpoint of the corresponding seed and the same learning rate and schedule as Specht Triple, and no control hyperparameter sweep was performed.

The replication unit is the seed. Reductions are ratios of paired means, bootstrap intervals resample the ten seeds jointly, and the exact sign-flip test enumerates all 2^{10} sign assignments. Sign-flip tests are reported one-sided, in the hypothesised direction, with the two-sided value as a sensitivity analysis. The comparison of Specht Triple with Projector Control is the designated primary test and is unadjusted; the comparison with Control Triple and the taskwise and subgroup analyses are secondary. Table A.2 gives the per-seed scores.

Seed	GIN	Control Triple	Projector Control	Specht Triple	Reduction
11	0.09307	0.08139	0.08151	0.06383	21.69%
23	0.09019	0.07012	0.05813	0.06492	−11.68%
37	0.08733	0.07603	0.08714	0.06515	25.24%
53	0.09374	0.07637	0.06541	0.06288	3.87%
71	0.08861	0.07254	0.07917	0.06023	23.92%
89	0.09002	0.07533	0.05430	0.05728	−5.49%
107	0.09424	0.07549	0.06422	0.06090	5.17%
127	0.09453	0.08012	0.07211	0.06220	13.74%
149	0.08920	0.08068	0.07899	0.06224	21.21%
173	0.09225	0.07590	0.05274	0.05742	−8.87%
Mean	0.09132	0.07640	0.06937	0.06171	11.05%

Table A.2. Mean standardised test MSE by seed. The last column is the reduction of Specht Triple relative to Projector Control; in the mean row it is the ratio-of-means reduction.

For the primary comparison with Projector Control, the absolute reduction in mean MSE is 0.00767 and the paired effect size (Cohen's d_z) is $d_z = 0.713$. For the secondary comparison with Control Triple, the paired effect size is $d_z = 3.38$, and removing any single seed leaves a reduction between 18.67% and 20.43%. Control Triple improves on GIN by 16.34% (10/10 seeds, interval [14.07%, 18.49%]). The taskwise reductions of Specht Triple relative to Control Triple win 10/10 seeds for diameter and 9/10 for algebraic connectivity. By family,

the reduction relative to Control Triple is about 10.6% (Barabási–Albert), 17.2% (Erdős–Rényi), 17.0% (two-block) and 22.4% (Watts–Strogatz), and the average error is lower for every order $n = 15, \dots, 24$. Under 20 redraws of the random identifiers on 512 held-out graphs, the standard deviation of predictions, in units of the training-target scale, is about 0.050 for diameter and 0.077 for algebraic connectivity.

A4. Computational cost

Inference was timed on the full test set with two warm-up and nine timed passes per seed (median recorded); training figures are from the original runs and are incremental to the frozen backbone.

	GIN	Control Triple	Specht Triple
Incremental training time (s)	–	30.09 ± 0.38	147.35 ± 2.53
Peak training memory (GiB)	–	0.167 ± 0.001	0.679 ± 0.001
Test-set inference time (s)	0.0291 ± 0.0007	0.0508 ± 0.0011	0.2042 ± 0.0073
Peak inference memory (GiB)	0.0162 ± 0.0002	0.0947	0.5924

Table A.3. Computational cost, mean ± standard deviation over the ten seeds. No matched timing was recorded for Projector Control.

Appendix B

Supplementary results for subset representations

This appendix collects the representation-stability discussion and character calculations supporting Chapter 7.

B1. Representation stability

The decomposition of Theorem 7.1.3 is into orbit spans, not generally into irreducible representations. Each summand records one overlap pattern among the input subsets. The indexing set stabilises for $n \geq rd$, although the modules $P_{\mathscr{H}}(n)$ themselves still depend on n. The irreducible constituents stabilise as well: since, for $S_m \leq S_n$ the pointwise stabiliser of $n-m$ points, $(S^\lambda)^{S_m} \neq 0$ if and only if $\lambda_1 \geq m$, every constituent S^λ of $\mathrm{Sym}^d V_{n,r}$ satisfies $n - \lambda_1 \leq rd$, and the multiplicity of each padded Specht module $S^{(n-|\nu|,\nu)}$ is eventually constant in n, because $n \mapsto \mathrm{Sym}^d V_{n,r}$ is a finitely generated FI-module [CEF15].

Symmetric powers of permutation modules have also been studied in the modular setting [Jia21]. Here the orbit description is useful because it keeps the overlap data visible without first decomposing into irreducibles. For example, at $n = rd$ the disjoint-edge summand is the Foulkes module $\mathrm{Ind}_{S_r \wr S_d}^{S_{rd}} \mathbf{1}$, whose general irreducible decomposition is an open problem [PW16].

B2. Character computations

The dimension formula (9) can also be evaluated from characters, as follows.

Remark B2.1 (Molien–Pólya formula). By Burnside's lemma, evaluating (9) requires only the cycle structures of the automorphism groups. Alternatively, one can sum over cycle types of S_n. For $g \in S_n$, a subset is fixed by g exactly when it is a union of cycles, so $f_r(g) := \left|X_{n,r}^{g}\right| = [z^r]\prod_c (1+z^{|c|})$, the product being over the cycles of g on $[n]$ and $[z^r]$ denoting the coefficient of z^r. These are the power-map characters of $V_{n,r}$, and the exponential form of Proposition 4.3.1 gives

$$\dim \operatorname{Hom}_{S_n}(\operatorname{Sym}^d V_{n,r}, V_{n,s}) = \frac{1}{n!}\sum_{g\in S_n} h_d(g) f_s(g), \qquad \sum_{d\ge 0} h_d(g)\,t^d = \exp\Bigg(\sum_{m\ge 1}\frac{t^m}{m} f_r(g^m)\Bigg),$$

where $h_d(g) = \chi_{\operatorname{Sym}^d V_{n,r}}(g)$ is computed by the recursion $d\,h_d(g) = \sum_{m=1}^{d} f_r(g^m)\,h_{d-m}(g)$. Replacing f_s by an irreducible character gives the multiplicities of Specht modules in $\operatorname{Sym}^d V_{n,r}$. Both sums depend on g only through its cycle type.

Appendix C

Projector entries and route calculations

This appendix supplies the explicit Johnson-projector entries and the proof of Proposition 8.2.2. These calculations justify the route set used in Section 8.3.

C1. Entries of the triple projectors

The eigenvalue of A_t on L_i is the Eberlein polynomial

$$P_t(i) = \sum_{h=0}^{t} (-1)^h \binom{i}{h}\binom{3-i}{t-h}\binom{n-3-i}{t-h},$$

and the valency of A_t is $v_t = P_t(0) = \binom{3}{t}\binom{n-3}{t}$ [BCN89, Section 9.1]. The entries of E_i are $(E_i)_{TT'} = \frac{d_i}{N}\frac{P_t(i)}{v_t}$ for $\left|T \setminus T'\right| = t$, by the reciprocity between the eigenvalues and dual eigenvalues of the scheme [Del73; BCN89]; they are listed in Table C.1.

$N\,(E_i)_{TT'}$	$t=0$	$t=1$	$t=2$	$t=3$
$i=0$	1	1	1	1
$i=1$	$n-1$	$\frac{(n-1)(2n-9)}{3(n-3)}$	$\frac{(n-1)(n-9)}{3(n-3)}$	$-\frac{3(n-1)}{n-3}$
$i=2$	$\frac{n(n-3)}{2}$	$\frac{n(n-7)}{6}$	$-\frac{n(2n-11)}{3(n-4)}$	$\frac{3n}{n-4}$
$i=3$	$\frac{n(n-1)(n-5)}{6}$	$-\frac{n(n-1)(n-5)}{6(n-3)}$	$\frac{n(n-1)(n-5)}{3(n-3)(n-4)}$	$-\frac{n(n-1)}{(n-3)(n-4)}$

Table C.1. Entries of the Johnson projectors of $V_{n,3}$, $n \geq 6$, as functions of $t = \left|T \setminus T'\right|$. The valencies are $v_t = 1,\ 3(n-3),\ \frac{3}{2}(n-3)(n-4),\ \frac{1}{6}(n-3)(n-4)(n-5)$.

C2. Proof of Proposition 8.2.2

Proof of Proposition 8.2.2. By Lemma 8.1.1 it suffices to decide, for $1 \leq i \leq 3$ and $j \notin \{0, i\}$, whether some even-degree coefficient $\kappa_j^{(i)}(2r)$ is positive. Evaluating (13) with Table C.1 gives

$$\kappa_2^{(1)}(2) = \frac{8(n-4)}{(n-2)^2(n-3)^2}, \qquad \kappa_3^{(1)}(2) = 0, \qquad \kappa_3^{(1)}(4) = \frac{576(n-6)}{n(n-2)^4(n-3)^4},$$
$$\kappa_1^{(2)}(2) = \frac{2(n-6)^2}{(n-2)^2(n-4)}, \qquad \kappa_3^{(2)}(2) = \frac{6(n-6)}{(n-2)(n-4)^2},$$
$$\kappa_1^{(3)}(2) = \frac{(n-5)(n-6)}{(n-3)(n-4)}, \qquad \kappa_2^{(3)}(2) = \frac{(n-5)\,(n^3-14n^2+63n-86)}{(n-3)^2(n-4)^2}.$$

For $n \geq 7$ all of these except $\kappa_3^{(1)}(2)$ are positive; for the last one, substituting $n = 7+m$ gives $n^3 - 14n^2 + 63n - 86 = m^3 + 7m^2 + 14m + 12 > 0$. Hence, by Lemma 8.1.1, every route (i, j) with $i \geq 1$ is realised, in degree two when $\kappa_j^{(i)}(2) > 0$, and in degree four for $(1, 3)$. Together with $(0, 0)$ and the absence of routes out of L_0 to other targets, this proves (a).

For (b), let $n = 6$. Complementation $T \mapsto [6] \setminus T$ is a permutation of the triples commuting with S_6, with matrix A_3. By the formula for $P_3(i)$, only the term $h = i$ survives when $n = 6$, so A_3 acts on L_i by $(-1)^i$. Because A_3 permutes coordinates, $A_3 f_{\mathcal{B}}(z) = f_{\mathcal{B}}(A_3 z)$ for every function $f : \mathbb{R} \to \mathbb{R}$. If $x \in L_2$, then $A_3 x = x$, so $f_{\mathcal{B}}(x)$ lies in the $(+1)$-eigenspace $L_0 \oplus L_2$ for *every* activation f; hence the routes $L_2 \to L_1$ and $L_2 \to L_3$ vanish. If $x \in L_1 \oplus L_3$, then $A_3 x = -x$, and writing $\sigma(z) = \lambda z + \mu\,|z|$ gives $p_j \sigma_{\mathcal{B}}(x) = \lambda p_j x + \mu\, p_j\, |x|_{\mathcal{B}}$, where $|x|_{\mathcal{B}} = \sum_{b \in \mathcal{B}} |x_b|$ b is the coordinatewise absolute value, which is fixed by A_3 and therefore lies in $L_0 \oplus L_2$. For $x \in L_1$ and $j = 3$, or $x \in L_3$ and $j = 1$, both terms vanish. The remaining routes $(1, 2)$ and $(3, 2)$ are realised by Lemma 8.1.1, since $\kappa_2^{(1)}(2) = \kappa_2^{(3)}(2) = \frac{1}{9}$ at $n = 6$.

Finally, for σ_c with $c \neq 0$, Proposition 6.2.2 gives $a_3(\sigma_c; 1) = \frac{c}{6}\varphi(c) \neq 0$, and (13) gives $\kappa_3^{(1)}(3) = 48/((n-2)(n-3))^3$ and, at $n = 6$, $\kappa_1^{(3)}(3) = \frac{1}{36}$. Both are positive, so by (10) both routes acquire a nonzero third-chaos component. □

A complementary symmetry-level inventory, namely the multiplicities $\dim \mathrm{Hom}_{S_n}(\mathrm{Sym}^k L_i, L_j)$ for small k, can be computed from characters using Remark B2.1 and the character formula $\chi^{(n-i,i)} = \chi_{V_{n,i}} - \chi_{V_{n,i-1}}$ implied by Proposition 8.2.1; tables for $n = 20$ are given in [Sha26b].

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